
UNIT 7 DEMAND FORECASTING

Objectives

After Finishing this unit, you should be able to:

- Acquaint with the concept of forecasting.
- Appreciate nature and the need for forecasting.
- Know the various steps involved in the forecasting process.
- Find out the best way to select an appropriate forecasting technique.
- Develop familiarity with different Qualitative Techniques.
- Learn different Quantitative Forecasting Techniques.
- Understand the applicability of forecasting in production/operations.
- Learning the importance of forecasting in decision-making.
- Using forecasting techniques in operations management.

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7.1 INTRODUCTION

Forecasting is the art and science of predicting future events; until the last few decades, forecasting was mainly an art, but it has also become a science. There will be no universal forecasting method for all situations. Generally, people forecast and invest in saving for a better future and take care of contingencies like severe illness and unlikely heavy expenses that may or may not occur to protect their families from the affected or to reduce the impact of such a situation. Similar uncertain problems will face decision-makers in their respective fields because of uncertainties in markets and the business environment.

Assume that a manager decided to set up another plant to produce more products to meet the very high demand existing in the present market. However, it took much time to start production; by then, the earlier planned demand no longer existed. Thus, when there is time to implement the plan, everyone likes to know the nature of events in the future to act accordingly and plan to the best of his/her ability, judgment, and experience.

Forecasts rarely exactly match the future as little variations are often observed due to extra capacity, inventory, or rescheduling of orders. However, significant variations from forecasts can create chaos in business operations. Therefore, the quality of the decision depends mainly on the quality of the forecast, and the plan's effectiveness depends upon the accuracy level in assessing future events.

7.1.1 Concepts of Forecasting

Generally, forecasting and predicting are used interchangeably in day-to-day usage. Adam and Ebert have defined and distinguished the two as follows:

Forecasting: "Is a process of estimating a future event by casting forward past data. The past data are systematically combined in a predetermined way to obtain the estimate of the future".

Prediction: "Is a process of estimating a future event based on subjective considerations other than just past data; these subjective considerations need not be combined in a predetermined way."

Thus, forecast estimations of events may be the demand for a product, rainfall at a particular place, the population of a country, or the growth of technology, etc. The forecast value is only an estimate based on past historical data related to a particular event, for which proper care must be given during estimation.

7.1.2 Nature of the Forecast

Forecasting typically involves predicting future values of variables and alerting people in advance to upcoming events. Forecasting is crucial in a company's decision-making process and can be done for long- or short-term goals. Forecasting is the first level of decision activity in an industrial organization that determines the demand for a specific product before making

additional decisions like material planning, scheduling, production method (mass or batch production), implementation, etc.

Forecasting serves as the foundation for corporate planning coordination. Based on the anticipated value, a functional manager judges activities crucial to the organization. The estimation of predicted values of particular specified (single or multiple) indicators linked to a decision on a planning scenario should therefore be done with due caution. The quantity of indicators depends on how the prediction will be used.

7.1.3 Need for Forecast

Two fundamental reasons for forecasting in any area, specifically production or operations management, are as follows.

1. Any decision regarding a present action is devised to take care of some contingency situation in the future or towards a purpose/target to be achieved to take advantage of or minimize the impact of the foreseen conditions.
2. Time is a resource that is needed for the planning, organization, and execution of a plan. While some situations call for very little time, others call for many years. Therefore, appropriate actions can be planned and implemented "in time" if a future forecast is available. In business, forecasts can be classified into:
 - A. Technology forecasts:** According to the technology prediction, the technology combines hardware and software. Technology forecasting deals with specific traits like technical performance level and rate of technological advancement. Technological forecasting entails predicting the qualities of future practical tools, products, methods, and approaches.
 - B. Economic forecasts:** Businesses and government organizations gather data and make educated guesses about the general business environment. These estimates help government organizations forecast future tax receipts, levels of business growth, employment, inflation, etc. and help business circles plan their future activities based on those levels of growth.
 - C. Demand forecast:** It provides the anticipated demand for goods or services, which is the fundamental input for corporate planning and control—affecting every choice a corporate house makes.

Table 7.1: Factors affecting demand Forecasting

Business life Cycle	Random Variable	Customers plan	Product life cycle
Competitors' efforts and prices	Customer's confidence and attitude	Quality	Credit policy
Design of goods or services	Reputation for service	Advertising	Sales effort

7.1.4 General Steps in the Forecasting Process

The general steps in the forecasting process are as follows:

- 1) Identify the needs.
- 2) Select the Period (Time Horizon) of Forecast: Its general estimation of the time taken to implement the plan and rough estimation of how long it will be implemented.
- 3) Select the Indicators Relevant to the Need: Based on the product or product line of the business Indicator must be Identified. Some indicators could be Industry Sales, present and projected capacity of Competitors, Population projection, Income levels, Economic development, etc.
- 4) Select the Forecast Model to be Used: One should have proper knowledge regarding various forecasting models available and when to apply these models (Appropriate situation) based on reliability and data available.
- 5) Data Collection: Collect the data regarding Identified Indicators from appropriate sources to be compatible with the forecasting model. Make sure to collect data from appropriate periods in the past to have proper projections.
- 6) Prepare Forecast: Apply the model using the data collected and get the statistical values of the forecast.
- 7) Evaluate the Forecast: As the values obtained in the forecast cannot be used blindly, they are evaluated in terms of feasibility, probability of occurrence, against some related phenomenon of variables, etc., to get the final forecast.

7.1.5 Source of Data for Forecasting

The data for forecasting is vital, but it can be like Garbage-in and Garbage-out without knowing how to interpret the relevant data and picking a suitable model for the forecast. Authenticity can be maintained if the data is obtained from government publications, company records, newspapers, journals, surveys, etc. Since the forecasting is done from past data, the past time period during which the data can be collected becomes crucial. If the time period is taken, the reliability of forecasting increases along with the cost and time of data collection. Hence, a suitable time period should be preferred.

7.1.6 Selection of a Suitable Forecasting Technique

Selecting a suitable forecasting technique depends on the following factors that companies consider essential:

1. **User Sophistication:** Forecasting methods must be matched to the knowledge and sophistication of the manager (user). If not, managers could be reluctant to use results from techniques as they need help understanding.
2. **Software Sophistication vs. ease of application:** The status of current forecasting systems in use should match the advancements in the field.

3. **Time and Resources available:** The forecasting method is selected based on the time available to collect the data and prepare the forecast, the necessary resources involved, and the costs of the forecasting method. Detailed forecast requires actual data to be collected and takes several months, costing thousands of money. At the same time, routine forecasts made by computerized systems cost moderately and require less time.
4. **Use or Decision characteristics:** The forecasting method should be related to decisions in terms of accuracy, time horizon, and number of items to be forecast.
5. **Historical Data Availability:** Selecting an appropriate forecasting method is often constrained by data availability and the level of detail required.
6. **Data Pattern:** The data pattern is the most critical factor affecting the forecasting method and determining whether a time-series method or a casual model is needed. If the time series is flat, the first-order method can be selected, but if the data show trends or seasonal patterns, an advanced method needs to be used. A qualitative method may be selected if the data pattern is unstable over time.
7. **The best fit to historical data:** The predictions should fit accurately in the future. Usually, it is said that the model that fits best for historical data is the best predictive model with minor errors, but this may only be true in some cases.
8. **Qualitative forecasting versus quantitative forecasting:** It has always been a concern of managers. Quantitative forecasting models do better when the data includes significant random noise or non-linear seasonal patterns. While qualitative models outperform when simple (first order), exponential models with an arbitrary value are used.

7.2 FORECASTING MODELS

Forecasting can be done for "longer Range," a period of two years or more, or "medium range," a period of six months to two years into the future. While medium-term planning is the typical time frame for general decisions like aggregate planning, budgeting, and other resource acquisition and allocation decisions, long-term forecasting is a common horizon for planning facilities and processes. The foresight There are two categories of models or procedures: qualitative techniques and quantitative techniques. At the same time, time series and causal forecasts are additional categories for quantitative techniques.

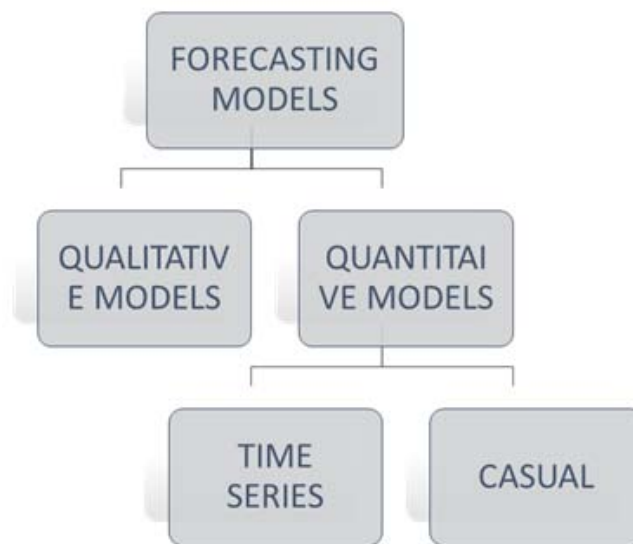


Figure 7.1: Forecasting Models

7.2.1 Qualitative Forecasting Techniques

Qualitative forecasting methods depend on judgments given by management, and they do not use any specific models or processes. Thus, this method is also known as Judgement Method. This technique is subjective as different individuals using the same qualitative method derive different results and arrive at different forecasts. This method is most useful when there is a lack of data or when past data are not a reliable predictor of the future. Forecasting decision is made utilising managerial judgments from the best available relevant data using their experience and implicit simple mathematical model in this technique. Hence, objective data or subjective impressions are taken as inputs are translated into Forecasts in the human mind with the support of formal analysis.

Qualitative techniques are used in Events like New product development, Technology development & innovation, the arrival of a new competitor, or formation of a new coalition, or calling a labour strike since Historical data is unavailable. It is often used for long-range and Middle range forecasts where external factors may play a significant role. Thus, it may show Unstable patterns. While utilizing these techniques, a Forecasting expert should be able to sketch the general direction of future developments, anticipate major points where developments will hinge, and make contingency forecasts for alternatives.

Some Qualitative or Judgement techniques are listed below:

1. Interviews – Direct, Telephonic, or Online
2. Traditional or Structured Meetings
3. Role Playing
4. Questionnaire – In person or Mail
5. Delphi Technique
6. The Market Survey and Market Testing

Qualitative methods are used primarily for long and medium-range forecasting involving process design and facilities planning.

Delphi technique:

Delphi Technique is an “Opinion – Capture Technique” type where experts' viewpoints are collected to analyze, and recommendations are generated as solutions for problems or issues. Thus, it is a method of forecasting applied to the subjective nature of demand values. This method is used when estimating the demand becomes difficult, and there are mixed reactions in the market regarding a particular thing. Example: technology forecasting has no quantitative data based on which the future technology can be predicted, so information is collected regarding advancements in technology in the form of opinions from panel members who are knowledgeable persons in that area. From these opinions, a consensus is arrived at as a conclusion.

The precautions to be followed in this method are as follows.

- Panel members must be unknown to each other.
- The initial questionnaire should be unambiguous and explain every matter about which opinion is sought.
- Opportunity to review views given by any panellist.
- Statistical assessment of the group response.
- Feedback from individuals and groups is to be controlled, and contributions of information and knowledge are discussed with all panellists again.

Then the panellists' opinions are evaluated for similarity, and if there are significant differences, a further round of distribution among the members is required without mentioning the names of the individuals who supplied the comments. They will be asked to reevaluate their positions if their opinions considerably depart from the middle 50% of the estimate. However, if they choose to maintain their original viewpoints, they will be required to defend them. The Delphi method is thus an iterative procedure until the panel reaches an agreement on the subject at hand or converges on a particular value or a range of values as indicated by the desired precision. Criticisms that emerged against Delphi are:

1. The methodology needs a theoretical foundation.
2. Not goal specific.
3. Deviation may result from variations in experts' subjective definitions.
4. The experts' occasional use of random samples
5. incorporating value judgments
6. Content and construct validity, as well as unmeasured reliability.

Despite these criticisms, the Delphi method is still widely used, accepted, and preferred, especially when a group consensus is required. Because of the following advantages:

1. There is no excessive influence from strong or persuasive individuals.

2. He does not have to disagree publicly with eminent figures.
3. There is no temptation to get carried away by popular opinion.
4. Because his identity is protected, he can always change his mind.
5. Diverse viewpoints are also consciously gathered. Relatively cheap to administer and economizes on the time required by members.
6. Facilitates conceptualizations of complex phenomena.

Market survey and market testing:

New products and services are often subjected to extensive market research before a final decision is made on introducing the new output. Devices such as mail questionnaires, telephone surveys, consumer panels, and test markets are used to ascertain estimates of value, perceived quality, and demand. Test marketing can also provide data over a long period to track changes in shopping and/or buying trends (called longitudinal research). A test market is some specific geographical region selected because it represents some segment of the organization's overall market for the new output. The test market requires the product or service to be introduced into the limited test area. Test marketing provides management with information about actual consumer behavior rather than simply consumer attitudes, opinions, or intentions since there are often significant differences between a consumer's attitudes and his/her actions. As with consumer market surveys, it is also possible to conduct industrial market surveys. Here, the subjects are vendors, purchasing agents, and other more knowledgeable individuals than consumers.

7.2.2 Quantitative Forecasting Techniques

Quantitative methods use some or the other underlying model to arrive at a forecast, assuming that past historical data and its patterns are reliable sources for future predictions upon being processed by time series or causal mode. These methods are more accurate, and computers can speed up the process. Quantitative Forecasting Techniques can be classified as Time series models or Causal models.

1. **Time Series models:** Time series models relate quantities strictly to time. They are time-ordered sequences of observations taken at regular intervals over a period. This model is further classified as follows:
 - a. Trend projection(long-term)
 - Trend Projection using the Least Square Method
 - b. Decomposition methods(intermediate term)
 - Simple moving average
 - Double moving average
 - Semi-average method
 - c. Smoothing methods(short-term)
 - Single Exponential smoothing

- Double Exponential smoothing
2. **Causal models:** Casual Models relate quantities to other factors.

This model's future consists of

- Simple Regression analysis
- Multiple Regression analysis
- Correlation analysis
- Econometric model
- Input-Output model
- Simulation method

Trend projection using the least square method: The trend can be a straight line or a curvilinear one. In Trends, we usually study historical patterns to project these past patterns into the future to avoid unnecessary trend components that could be disadvantageous.

Linear trend – the objective is to find the value of slope and Intercept from given independent variable values so that the general trend is known using a straight line.

$$Y_t = b_0 + bx_t$$

where Y_t = Forecast for period t ,

b = slope of the line,

b_0 = Y Intercept

t = dependent variable estimated value

x = independent variable

However, we use the Least Square Method to find the best-fitting line.

Least Square Method: The process of finding the best-fitting curve for a set of data by reducing the sum of squares of all the deviations from the given offset points must be minimum. While finding the relationship between two variables, the trend of outcomes is estimated quantitatively.

Let us assume,

Given data points are $(x_1, y_1), (x_2, y_2), (x_3, y_3), \dots, (x_n, y_n)$.

Where all x 's are independent variables,

all y 's are dependent ones,

$f(x)$ is the fitting curve.

d is an error or deviation from each given point. $d_n = y_n - f(x_n)$

sum of squares of all the deviations = S , should be minimum

Suppose when we have to determine the equation of the line of best fit

for the given data,

1. Use the equation of least square line as given by $Y = a + bX$
2. Normal equation for 'a': $\sum Y = na + b\sum X$
3. The normal equation for 'b': $\sum XY = a\sum X + b\sum X^2$
4. Solving these two normal equations and getting the trend line equation with a line of best fit with the formula $y = ax + b$

Example 1: Consider the time series data below and use the least square method to determine the equation of the line of best fit for the data.

x	13	8	7	15	16	8	11	10	11	13
y	9	17	6	17	14	9	14	11	6	19

Solution:

Mean of x_i values = $(13+8+7+15+16+8+11+10+11+13)/10 = 112/10 = 11.2$

Mean of y_i values = $(9+17+6+17+14+9+14+11+6+19)/10 = 122/10 = 12.2$

The straight line equation is $y = a + bx$.

The normal equations are $\sum y = na + b\sum x$

$\sum xy = a\sum x + b\sum x^2$

x	Y	x^2	xy
13	9	169	117
8	17	64	136
7	6	49	42
15	17	225	255
16	14	256	224
8	9	64	72
11	14	121	154
10	11	100	110
11	6	121	66
13	19	169	247
$\sum x = 112$	$\sum y = 122$	$\sum x^2 = 1338$	$\sum xy = 1423$

Substituting these values in the normal equations,

$$10a + 112b = 122 \dots (1)$$

$$112a + 1338b = 1423 \dots (2)$$

Equation (1) $\times$ 112 – Equation (2) $\times$ 10,

$$1120a + 12544b - (1120a + 13380b) = 13664 - 14230$$

$$-836b = -566$$

$$b = 566/836$$

$$b = 0.677$$

Substituting $b = 0.677$ in equation (1),

$$10a + 112(0.677) = 122$$

$$10a + 75.824 = 122$$

$$10a = 122 - 75.824$$

$$10a = 46.176$$

$$a = 46.176/10$$

$$a = 4.6176$$

Therefore, $y = a + bx$ becomes the required trend line equation.

$$y = 4.6 + 0.68x$$

Now, we can find the sum of squares of deviations from the obtained values as:

$$d_1 = [9 - (4.6 + 0.68 \cdot 13)] = -4.44$$

$$d_2 = [17 - (4.6 + 0.68 \cdot 8)] = 6.96$$

$$d_3 = [6 - (4.6 + 0.68 \cdot 7)] = -3.36$$

$$d_4 = [17 - (4.6 + 0.68 \cdot 15)] = 2.2$$

$$d_5 = [14 - (4.6 + 0.68 \cdot 16)] = -1.48$$

$$d_6 = [9 - (4.6 + 0.68 \cdot 8)] = -1.04$$

$$d_7 = [14 - (4.6 + 0.68 \cdot 11)] = 1.92$$

$$d_8 = [11 - (4.6 + 0.68 \cdot 10)] = -0.4$$

$$d_9 = [6 - (4.6 + 0.68 \cdot 11)] = -6.08$$

$$d_{10} = [19 - (4.6 + 0.678 \cdot 13)] = 5.56$$

$$\sum d^2 = (-4.44)^2 + (6.96)^2 + (-3.36)^2 + (2.2)^2 + (-1.48)^2 + (-1.04)^2 + (1.92)^2 + (-0.4)^2 + (-6.08)^2 + (5.56)^2 = 159.28$$

Simple Moving Averages: A simple moving average is a method of computing the average of a specified number of the most recent data values in a series. The formula to compute the simple moving average is as follows:

$$M_t = \frac{1}{n} \{D_{t-(n-1)} + D_{t-(n-2)} + \dots + D_{t-2} + D_{t-1} + D_t\}$$

Where M_t = Simple moving average at the end of period t (used as a forecast for period $t+1$)

D_t = Actual demand in period t

n = number of periods included in each average.

Example 2: Calculate the three months moving average for the following monthly demand and forecast demand for period 13.

Time(month)	1	2	3	4	5	6	7	8	9	10	11	12
Demand	100	105	92	128	95	101	80	83	111	109	94	88

Solution:

Time (t)	Monthly demand (t)	Moving Average M(t)	Forecast (Ft)	Error (et)
1	100			
2	105			
3	92	99		
4	128	108.33	99	29
5	95	105	108.33	-13.33
6	101	108	105	-4
7	80	92	108	-28
8	83	88	92	-9
9	111	91.33	88	23
10	109	101	91.33	17.66
11	94	104.66	101	-7
12	88	97	104.6667	-16.66

Mean Forecast Error (MFE) = - 0.92

The values shown under column M(t) are computed as:

Moving Average Period (n) = 3

$$M(3) = \frac{100+105+92}{3} = 99$$

$$M(4) = \frac{105+92+128}{3} = 108.33$$

The forecast for period 13 is: F (13) = M (12) = 97

Comments:

- In most cases, this method is applied to forecast for only one period into the future.
- The forecaster must wait until demand entries are available before making the first forecast.
- This method applies only to horizontal demand data patterns.
- This model requires storing the first n observed values, which consumes a considerable amount of computer storage space if the demands of several products are to be forecasted.
- A more significant smoothing effect could be obtained by including more observations in the moving average.

Weighted moving average: Equal weights were assigned to all periods in the computation of the simple moving average. The weighted moving average assigns more weight to some demand values (usually the more recent ones) than others.

Example 3: The computation for three months weighted moving average with a weight of 0.5 assigned to the most recent demand value, a weight of 0.40 assigned to the next most recent value, and a weight of 0.10 assigned to the oldest of the demand value included in the average.

Time (t)	Monthly demand (t)	Moving Average M(t)	Forecast (Ft)
1	135		
2	145		
3	125	134	
4	155	142	134
5	125	137	142
6	145	138	137

$$\text{Weight MA}_3 = \frac{((0.1 \times 135) + (0.4 \times 145) + (0.5 \times 125))}{0.1 + 0.4 + 0.5} = 134$$

Formula:

Weight $\text{MA}_i = \text{Where } i = 1, 2, 3 \text{ if } \frac{\sum_{i=1}^n W_i D_i}{\sum_{i=1}^n W_i}$ three periods are moving average.

$i=3$ corresponds most recent time period

$i=1$ corresponds oldest time period.

W_t = weight for the time period t

In the above example $W_1 = 0.1$, $W_2 = 0.4$ and $W_3 = 0.5$

Semi-Averages Method: In this method, the whole data is divided into two parts to time; if we are given Y_t for t from 1991 to 2002, i.e., over 12 years, the two equal parts will be the data from 1991 to 1996 and 1997 to 2002. In case of an odd number of years, the two parts are obtained by discarding the value corresponding to the middle year; for example, for the data from 1992 to 2002, the two parts would be the values for 1992-96 and 1998-2002, the value corresponding to middle year viz., 1997 being discarded. Next, we compute the arithmetic mean for each part and plot these two averages against the mid-values of the respective periods covered by each part. The line obtained by joining these two points is the required trend.

Remarks:

- For an even number of years like 2, 4, 6, etc., the centering of an average of each part would create problems.

Example: for the data 1991 to 2002 ($n = 12$), let the two averages be X_1 (say) for the period 1991-1996 and X_2 (say) for the period 1997-2002. Here, X_1 will be plotted against the two mid-values, viz. 1993 and 1994 for the period 1991-1996 against 1st July 1993. Similarly, for 1997-2002, X_2 will be plotted against the two mid-values, viz. 1999 and 2000 against 1st July 1999.

- Compared with the graphic method, the apparent advantage of this method is its objectivity because everyone who applies it would get the same results.
- Compared to the 'method of least squares of the 'moving average method, it is readily comprehensible.
- Limitations: This method assumes a linear relationship between the plotted points, which may not exist. Moreover, the limitations of arithmetic mean as an average also stands in its way.
- The line may be extended both ways to eliminate intermediate or future values.

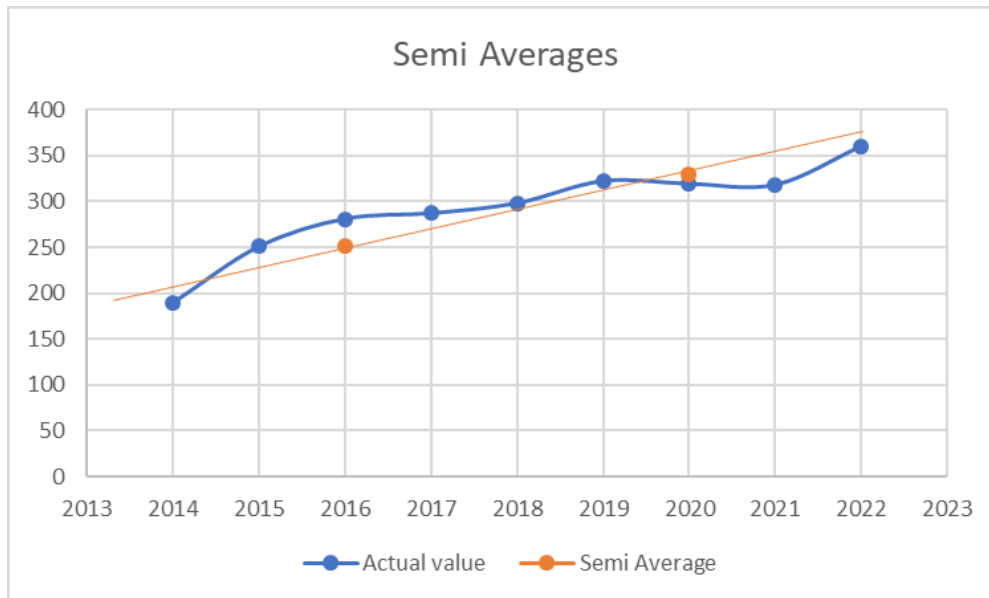
Example 4:

From the following series of annual sales data, find the trend line by the method of semi-averages.

Year	2014	2015	2016	2017	2018	2019	2020	2021	2022
Actual value	190	251	281	287	298	322	319	318	360

Solution: Here, the number of years is 9, i.e., odd. The two equal parts will be 2014 to 2017 and 2019 to 2022. The value for the middle year, viz., 2018, is ignored.

Year	Actual Value	4 Year Semi-Totals	Semi-Average
2014	190	1009	$1009/4=252.25 \approx 252$
2015	251		
2016	281		
2017	287		
2018	298	Ignored	
2019	322	1319	$1319/4=329.75 \approx 330$
2020	319		
2021	318		
2022	360		



The average value of the first half of the given data (for years 2014 to 2017) is 252 and will be plotted against the middle of the year 2015 and 2016, and the average value of the second half of the given data (for years 2019 to 2022) is 330 and will be plotted against the middle of the years 2020 and 2021. The average value of any group of observations lies at the center of the data set.

Single (simple) Exponential smoothing method: The exponential smoothing average is a weighted moving average. This method maintains a running average of demand. It modifies it proportionally for each period depending on the gap between the most recent actual demand figure and the current average value.

$$F_t = F_{t-1} +$$

Where $F_t =$ Smoothed average forecast for period t .

$F_{t-1} =$ previous period forecast.

$\alpha =$ Smoothing constant, the weight given to previous data ($0 \leq \alpha \leq 1$).

$D_{t-1} =$ previous period demand

If α is equal to 1, the latest forecast would equal the previous period's actual demand value. The preferred range α is from 0.1 to 0.3. The application of this technique is demonstrated in the following example.

Example 5: A firm uses simple exponential smoothing with $\alpha = 0.03$ to forecast demand. The forecast for the first week of January was 440 units, whereas the actual demand was 480 units.

Calculate (a) Forecast the demand for the next week of January.

(b) Assume the actual demand for the next week is 460. Forecast the demand up to February third/3rd February week, assuming the subsequent demands to be 465, 434, 420, 498, and 462 units.

Solution:

- a) The forecast for the next week of January is computed as shown below.

$$\begin{aligned} F_t &= F_{t-1} + \alpha (D_{t-1} - F_{t-1}) \\ &= 400 + 0.2(450 - 400) \\ &= 410 \text{ units.} \end{aligned}$$

- b) The workings for the remaining weeks are shown in tabular form of Smoothed Average Forecasts

Week	Demand D_{t-1}	Old Fore- cast (F_{t-1})	Forecast Error ($D_{t-1} - F_{t-1}$)	Correction $\alpha (D_{t-1} - F_{t-1})$	New Forecast (F_t) $F_{t-1} + \alpha (D_{t-1} - F_{t-1})$
Jan. Week 1	450	400	50	10	410
Week 2	460	410	50	10	420
Week 3	465	420	45	9	429
Week 4	434	429	5	1	430
Feb. Week 1	420	430	-10	-2	428
Week 2	498	428	70	14	442
Week 3	462	442	20	4	446

In the above Table, the initial forecast was available. If no previous forecast value is known, the 'old forecast' starting point may be estimated or taken to be an average of the values of some preceding periods.

Regression: Regression is the statistical process of estimating the value of a dependent variable, y, based on an independent variable, x. While multiple regression involves two or more independent variables, simple regression only uses one independent variable. The simple linear regression is:

$$y_c = a + bx$$

Where Y_c is the estimated demand,
x is the independent variable
a is y-intercept
b is the slope.

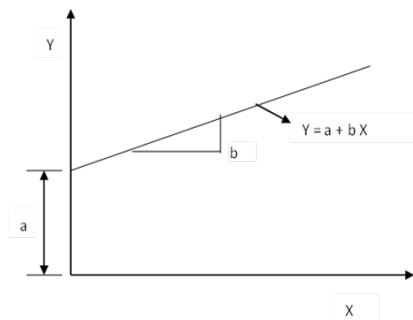


Figure 7.2: Regression

We assume that n pairs of x and y values have been observed for a simple linear regression model. We denote these pairs $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$. The symbol y denotes observed values of y , and y_c denotes points expressed by the equation $y_c = a + bx$.

The ' y ' values, which have been observed, do not fall exactly on the line because of random errors in the data. For each observed point, the error can be expressed as $y_c - y_i$, and the total variance or squared error due to all the points is then:

$$\Sigma (y_c - y)^2 = \Sigma (a + bx - y)^2$$

In regression analysis, the objective is to minimize the error of the above equation by choosing values of a and b . The formulas to compute the constants of the model are as follows.

$$a = \bar{Y} - b\bar{X}$$

$$b = \frac{n \sum xy - [\sum x \sum y]}{n \sum x^2 - (\sum x)^2}$$

For a given set of n data pairs, the values of a and b can be found in the equations. These values describe the straight line, the best (least squares) predictor of demand y based on the independent variable x .

Example 6: A company thinks its annual earnings depend on the amount it spends on research. Below is a list of data for the previous six years. When the cost is 6 units, calculate the profit.

Year	Expenditure for research (X)	Annual Profit (Y)
2014	12	60
2015	13	65
2016	15	74
2017	14	70
2018	21	80
2019	15	71
2020	16	?

Solution:

Year	X	Y	XY	X ²
2014	12	60	720	144
2015	13	65	845	169
2016	15	74	1110	225
2017	14	70	980	196
2018	21	80	1680	441
2019	15	71	1065	225
Total Σ	90	420	6400	1400

$$\bar{X} = \frac{90}{6} = 15$$

$$\bar{Y} = \frac{420}{6} = 70$$

$$b = \frac{((6 \times 6400) - (90 \times 420))}{((6 \times 1400) - (90 \times 90))} = \frac{n \sum xy - [\sum x \sum y]}{n \sum x^2 - (\sum x)^2} = 2$$

$$a = \bar{Y} - b\bar{X} = 70 - 2(15) = 40$$

The model is: $Y = a + bX = 40 + 2X$

The profit (y) when expenditure(x) is 6 units = $40 + 2 \times 6 = 52$ units of rupees

7.3 MEASURES OF FORECAST ACCURACY

The majority of decisions across all functions are influenced by demand forecasting. As a result, it needs to be approximated with the most significant degree of accuracy. Using some standard metrics to assess a forecasting technique's precision is unavoidable. This metric could represent the deviation between predicted values and actual demand or error.

The formula for error: $(et) = Dt - Ft$, where Dt = Demand for the period t.

Ft = Forecast demand for the period

et = Forecast error for the period t.

Types of errors

1. Mean Absolute Deviation (MAD)
2. Mean Square Error (MSE)
3. Mean Forecast Error (MFE)
4. Mean Absolute Percent Error (MAPE)

7.3.1 Mean Absolute Deviation (MAD)

It is the mean of absolute deviations of forecast demands from actual demand values. The MAD is sometimes called the mean absolute error (MAE).

$$\text{MAD} = \frac{\sum_{t=1}^n |D_t - F_t|}{n}$$

Where, D_t =

Actual demand for the period t

F_t = Forecast demand for the period t

n = number of time periods used.

7.3.2 Mean Square Error (MSE)

It is the mean of the squares of the forecasted demand's deviations from the actual demand levels. Minor errors typically have minimal effects on operations. Inventory or overtime labor may even out these mistakes. Even if there are only a few significant errors, having smoothed numbers for the forecast will be challenging. As a result, it is occasionally desirable to use an error measurement system that penalizes significant errors more severely than tiny errors. The mean square error (MSE) provides this type of forecast error measurement.

$$\frac{\sum_{t=1}^n (D_t - F_t)^2}{n}$$

MSE = Where, D_t = Actual demand for the period t

F_t = Forecast demand for the period t

n = number of time periods used.

7.3.3 Mean Forecast Error (MFE)

It is the mean of the deviations of the forecast demands from the actual demands.

$$\frac{\sum_{t=1}^n (D_t - F_t)}{n}$$

MFE = Where, D_t = Actual demand for the period t
 F_t = Forecast demand for the period t
 n = number of years (time period) used.

7.3.4 Mean Absolute Percentage Error (MAPE)

It is the mean of the percent deviations of the forecast demands from the actual demands.

Example 7

$$\frac{1}{n} \sum_{t=1}^n |D_t - F_t|$$

A forecast of 165 units was made for the demand in every period. The calculations of these errors are also shown in the table.

Solution:

Demand		Forecast	Deviation	Absolute Deviation	Squared Error	Percentage Error	Absolute % Error
T	D _t	F _t	D _t -F _t	D _t -F _t	(D _t -F _t) ²	$\frac{D_t - F_t}{D_t} \times 100$	$\frac{ D_t - F_t }{ D_t } \times 100$
1	130	155	-25	25	625	-19.2308	19.23077
2	140	155	-15	15	225	-10.7143	10.71429
3	145	155	-10	10	100	-6.89655	6.896552
4	155	155	0	0	0	0	0
5	160	155	5	5	25	3.125	3.125
Sum			+5	55	-975	55	39.96661

$$MAD = \frac{55}{5} = 11$$

$$MSE = \frac{975}{11} = 195$$

$$MAPE = \frac{39.966}{5} = 7.99\%$$

$$MF = \frac{-45}{5} = -9$$

7.4 FORECASTING PATTERNS IN BUSINESS

Forecasting is based on analyzing historical event patterns as they relate to time. Relationships between two or more variables make up another pattern. The historical data gathered allows for the immediate identification of such a trend. With the fundamental premise that demand can be broken down into components like average, level, trend, seasonality, cycle, and error, time series methods are used to analyze in-depth past demand patterns over time and project these patterns into the future. These elements will add to the original time series when added together (or, in some cases, multiplied).

Table 7.2: Different Trends

Horizontal	Fluctuation of data around a constant mean
Trend	Systematic increase or decrease in the mean of the series over time
Seasonal	A repeatable pattern of increases or decreases in demand, depending on the time of day, week, month, or season
Cyclical,	Less predictable gradual increases or decreases in demand over long periods of time (years or decades)
Random	Variation in demand that cannot be forecasted

1. Historical pattern (stationary pattern): This exists when the mean value does not change over time, and there is no trend in the data.
2. Seasonal demand pattern: A pattern occurs when the series varies in response to a seasonal element. The sale of refrigerators, soft drinks, wool goods, etc. may occur during the season, which can be months, quarters, weeks, etc.
3. Cyclic pattern: A cycle lasts no more than a year in a pattern of this kind. The cycle does not continually repeat itself. The finest examples are the cost of some metals, the gross domestic product, and other figures.
4. Trend: This kind of pattern is seen when the value of the variable changes over time, either upward or downward. Examples include the sales of various goods, stock prices, and business and economic indicators.

7.5 IMPORTANCE OF FORECASTING IN BUSINESS

The importance of forecasting lies in its ability to help planners(managers) take better actions for the future in advance while discharging their functions more effectively as they are better informed. This information helps them with focused thinking and generates choices toward set objectives. Thus, the importance of a forecast is directly proportional to the Results of an action based on a forecast & No forecast. The importance is greater if the difference is positive and significant; otherwise, the forecast is unimportant. This is the future impacted by the ability to use statistical forecasting techniques that are error-free in order to have less deviation and to generate reliable/accurate forecasts that are directly related. The more sophisticated forecast models often provide forecasts with smaller errors, but the cost of development of the model, forecasting, and maintenance tends to be high. Therefore, there has to be a tradeoff between the choice of model and the cost.

7.6 APPLICATIONS OF FORECASTING IN MANAGEMENT

Forecasting plays an essential role by providing inputs in various functional management areas such as Marketing, Production, Finance and Personnel.

The various varieties of forecasts used in each of the areas mentioned above are as follows:

Table 7.3: Applications of Forecasting in Management

Areas in Management	Forecast used
Marketing	• Demand forecasting of products • Forecast of market share • Forecasting trends in prices
Operations	• Process design • Capacity planning • Inventory control
Production	• Materials requirements • Trends in material and labor costs • Maintenance requirements • Plant capacity
Finance	• Cash flows • Rates of expenses • Financial planning • Revenues • Costs estimation • Tax planning
Personnel	• Number of workers in each category (Recruitment) • Labor turn-over • Absenteeism

7.7 APPLICATIONS OF FORECASTING IN OPERATIONS MANAGEMENT

Forecasting is such an increasingly valuable proposition for manufacturing companies that a study by Gartner (2016) indicated that forecasting (and the accuracy thereof) and demand variability were two of the most significant obstacles manufacturing companies encounter when overseeing their supply streams. Hence, forecasting is a critical value proposition for planning and production in today's modern manufacturing landscape.

Table 7.4: Application of Forecasting in Operations Management

S. No.	Applications	Some indicators	Timespan
1	New plant for product expansion	Long term demand Long term production The capacity of all producers	5 or more years or sometimes less

2	Seasonal Planning	Production	Demand cycles, Preferable from one peak to another peak	Usually 6 months- 2 years
3	Capital Planning		Long-term money market trends Interest rate trends	Usually, one or more years
4	Intermediate Planning	Operations	Intermediate demand shift in preferences	Up to 2 years
5	Short-run adjustment	production	Short-term demand Short-term material/ input protections	About 6 weeks
6	Scheduling	production	Order position, Emergency orders	Usually, 1 week
7	Product Development		Long-term sales trend specifically decline stage. Technological Developments in related fields	2 years and more depends on the usual time taken in product development

Table 7.5: Forecasting Methods Used in Operations Management

Forecasting in operations	Time horizon	Accuracy required	Number of products	Management level	Forecasting methods
Process design	Long	Medium	Single or few	Top	Qualitative and causal
Capacity planning Facilities	Long	Medium	Single or few	Top	Qualitative and causal
Aggregate planning	Medium	High	Few	Middle	Causal and time series
Scheduling	Short	Highest	Many	Lower	Time series
Inventory management	Short	Highest	Many	Lower	Time series

7.8 SUMMARY

The concept of forecast has been defined and differentiated from prediction. Forecasting deals with estimating future values of variables and letting us know in advance about the occurrence of future events. Prediction refers to subjective estimates of the future. A manager's experience and sound judgment are required for good predictions; statistical and management science techniques must often be used to make reasonable forecasts. Short-range demand forecasts determine the daily resource requirements needed for production, including labor and material, developing work schedules and shipping dates, and controlling inventory levels. Long-range forecasts are needed to plan new products for development and changes in existing products and to acquire plant, equipment, personnel, resources, etc. We also discussed the role played by the forecast in the day-to-day life of individuals and how it interests the manager in their job functioning. If a better/more accurate forecast is available, the manager can perform their job functions relating to planning, etc., quite efficiently.

7.9 SELF – ASSESSMENT EXERCISES

1. What is forecasting? Discuss various techniques of forecasting
2. What are the differences between qualitative and quantitative forecast methods? How often do organizations use forecasting techniques, and which techniques are used for what?
3. What do you understand by time series analysis? How would you conduct such an analysis for forecasting the sales of productivity of a firm?
4. What are the considerations in the choice of forecasting methods?
5. Explain the general process of forecasting.
6. Propound as to how forecasting can be helpful to production managers in their job functioning.
7. 6-Week demand and forecast data prepared by a forecasting method of a company's product is given below, made by a forecasting method being tested by the company.
 - a) Calculate the mean absolute deviation based on all the data.
 - b) calculate the mean squared error based on the data provided.

Week	1	2	3	4	5	6
Demand	146	183	146	175	194	177
Forecast	150	152	160	156	166	158

8. Sales data for a milk booth is provided in the following table in Liters (in hundreds). Forecast demand using Single Exponential Smoothing Method.

Month	1	2	3	4	5	6	7	8	9	10	11	12
Sales	192	202	188	210	204	194	195	211	202	107	298	195

Consider α 0.20 and an initial forecast of 180.00.

9. The sales of a company for 13 years are given below.

Year	1	2	3	4	5	6	7	8	9	10	11	12	13
Sales	92	114	88	105	105	116	125	124	128	152	171	186	215

- a) Fit simple regression for the data provided.
 - b) Forecast the sales for the 14th year.
10. Find 4 yearly Moving averages and comment.

Year	Income (Rs.)	Year	Income (Rs.)
2016	25,105	2009	44,890
2017	31,668	2010	66,979
2018	44,621	2011	70,000
2019	55,666	2012	78,043
2020	45,790	2013	66,921

2021	50,623	2014	75,889
2022	38,056	2015	69,032

7.10 FURTHER READINGS

Aswathappa K. and Shridhara Bhat K., Production and Operations Management, 1st ed, Himalayan Publishing House, Mumbai, 2015.

Lee J. Krajewski and Larry P. Ritzman, Operations Management – Strategy and Analysis, 6th Edn, Pearson Education, New Delhi, 2001.

Panneerselvam, R. Production and Operations Management, 3rd Ed. Prentice Hall of India, New Delhi, 2012.

Roberta S. Russel and Bernard W. Taylor III, Operations Management 7th Edn, Pearson Education, New Delhi, 2010

Schroeder, R. G., Goldstein, S. M., & Rungtusanatham, M. J. (2013). Operations management in the supply chain: Decisions and cases. McGraw-Hill/Irwin.



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UNIT 9 MANAGING DEMAND AND CAPACITY

Objectives

After studying this unit, you should be able to

- Explain the importance of managing demand and capacity for service organizations.
- Identify the demand patterns and their underlying reasons.
- Develop strategies for matching demand and capacity.
- Understand the concept of yield management.
- Provide strategies for managing customer waiting.

Structure

- 9.1 Introduction
- 9.2 Understanding Demand Patterns
- 9.3 Strategies for Matching Capacity and Demand
- 9.4 Yield Management
- 9.5 Managing Customer Waiting
- 9.6 Managing Demand and Waiting Lines : Case of an Amusement Park
- 9.7 Summary
- 9.8 Self Assessment Questions
- 9.9 Further Readings

9.1 INTRODUCTION

In the first unit of this course you have studied the characteristics of services which distinguish them from goods. One of the characteristics is perishability which means that services cannot be saved or stored. While marketers of physical goods hold inventories to buffer fluctuations in demand and supply, it is difficult or impossible for services marketers to do so. Therefore, many service businesses frequently find it difficult to match supply (capacity) and demand. At times there may be too much of demand (movie halls or restaurants on weekend evenings) and sometimes too little demand may exist (low weekend occupancies in business hotels). However, a theater owner or a restaurant cannot take an empty seat from Thursday night and add it to the capacity on Friday or Saturday night. A low occupation for a business hotel on weekends is an irretrievable loss. Similarly a hospital bed or an airline seat left vacant is a loss for ever. Inability to synchronize supply and demand has a significant impact on the service organization's bottom line through lost opportunity (when demand is greater than capacity) and through high costs (when demand is low in relation to fixed capacity resulting in under utilization of capacity)

Zeithaml, Parasuraman and Berry conducted an exhaustive survey in U.S.A (in 1980's) covering 1,000 service firms to find out the extent to which problems reported to be associated with services actually presented problems for the sample firms. Out of the eight commonly cited difficulties unique to services, only one problem area – 'The demand for services fluctuates' - received a mean score exceeding the mid-point on the 5 point scale (1 indicating no problem at all to 5 indicating a major problem). This problem received a score of 3.27 and another important finding was the absence of any significant difference across different types of services firms with regard to this problem.

Perception of demand fluctuation as a somewhat serious problem appeared to be universal. A similar study conducted in India in 1996-97 with reference to passenger airline industry covering more than a dozen airlines operating in India revealed similar results. The problem area that the demand fluctuates, received a mean score of 3.66 with 60 percent of the airlines indicating a score of 4 to 5. The next highest score was 2.94 with regard to problem area that service quality is difficult to control.

In this unit we will discuss various issues related to managing demand/capacity (supply) imbalances.

Activity 1

Compare a service organization with a manufacturing firm in terms of inventory capacity (e.g. car repair and maintenance service with an automobile manufacturer). Can you identify the implications for the service organization?

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9.2 UNDERSTANDING DEMAND PATTERNS

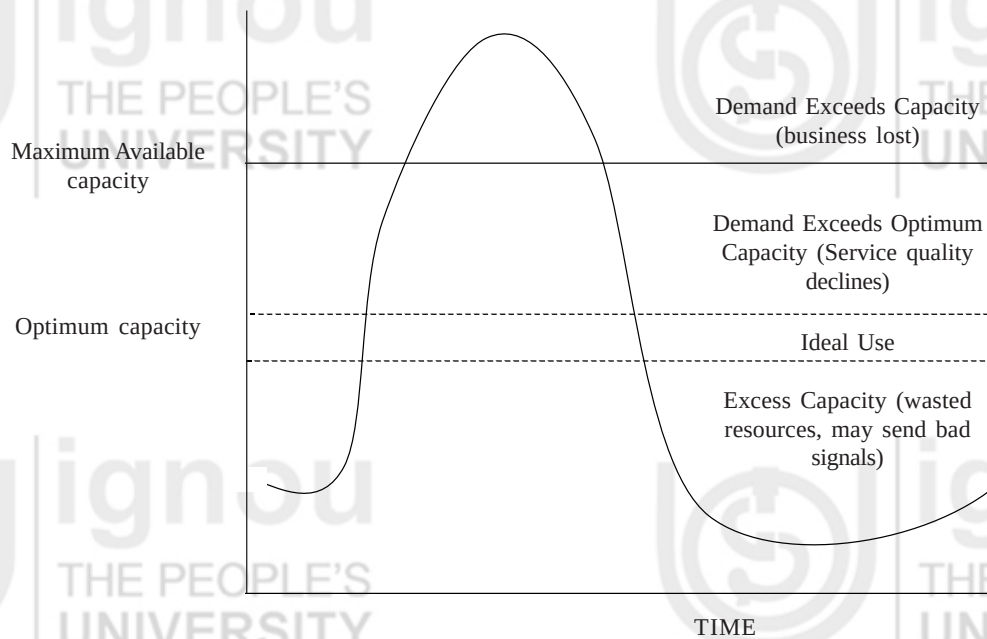
A service organization with a fixed capacity may be faced with one of the following four conditions:

1. **Excess demand:** The demand exceeds the maximum available capacity. This results in some customers being turned away. Also, even for the customers receiving the service, the quality of service may get affected (refer service delivery gap in the gaps model discussed in the previous unit). This may happen because of overcrowding and/or overstretching of resources.
2. **Demand exceeds the optimum capacity level:** Optimum capacity refers to the best use of capacity from the perspective of both, customers and the company. In most of the cases it is less than the maximum capacity. For example, in the counseling session at your study centre, while the maximum capacity of the rooms may be 60-70, the optimum capacity for conducting the session may be 30-40 only for ensuring proper interaction. In the situation when demand exceeds optimum capacity, while no one is turned away, customers may perceive deterioration in the quality of service delivered.
3. **Demand and supply are balanced at the optimum capacity:** This is the ideal situation. No one is turned away, no one is overworked in the staff and customers receives quality service.
4. **Excess capacity :** Demand is less than optimum capacity and therefore resources are underutilized. In certain cases this may also pose the risk that customers may have doubts about the service provider.

All the above four possibilities have been given in Fig. 9.1. The capacity of the service organization includes physical facilities, equipments and human resources.

The first step in finding out ways to manage demand and capacity is to understand the demand patterns and the factors which affect it. Better knowledge of demand patterns leads to better managerial decision making. The demand patterns may have a regular and predictable cycle or in some cases it may be largely random in nature and difficult to predict.

Fig 9.1: Implications of Cyclical Variations in Demand Relative to Capacity



(Source: Christopher Lovelock, Managing Services- Marketing Operations and Human Resources, 2nd Edn. Prentice Hall, p.155)

a) Predictable Demand Variations: Many businesses are subject to periodic cycles. These cycles may be daily (variations by hour), weekly (variation by day), monthly, seasonal and/or yearly. Try to identify services which may exhibit these types of predictable demand variations. For example, in case of a hotel there may be variation in demand on different days with business travelers going back on weekends, thereby reducing the demand. A hotel may also witness seasonal variation with high demand during a particular season due to large inflow of tourists. A restaurant faces hourly variations with low demand during 3.00 – 7.00 pm. (happy hours being offered by many restaurants during these times, with large price discounts). Amusement parks have greater demand during weekends as compared to weekdays and also greater demand is witnessed during school vacations.

As marketers you must find out if such predictable cycles exist in your business. If it is so, then you must find out and analyze the causes of these cyclical demand variations. Do these happen because of seasonal change, employment schedule, salary dates, school vacations, public or regional holidays etc? A proper analysis of these causes will help you in devising suitable strategy for managing the demand fluctuations.

b) Random Demand Fluctuations: At times the demand pattern may appear to be random with no apparently predictable cycle. Some degree of random variation in demand is faced by virtually all service firms. Although such variations cannot be predicted, marketers should nevertheless understand the underlying causes that typically cause them. For example bad weather may result in an unexpectedly low customer turnout at a movie hall or an amusement park. A disaster like accident, floods etc. may result in higher demand for health services and telecommunication services. A proper understanding of the underlying causes will prepare you to deal with such random demand fluctuations.

In order to understand the demand patterns and underlying causes, it may be useful to do a segmentation analysis. Different segments of customers may reveal different patterns as well as causes. Such an analysis will help you in pinpointing underlying causes of demand fluctuations and in identifying certain

Strategies of specific customer categories. This may help you in identifying certain segments which could be easily diverted to off-peak periods. You may also discourage segments that are not profitable or are inconsistent with the service image, at least during peak periods.

Once you properly understand the above issues you can suitably device strategies for matching demand and capacity.

Activity 2

Select a restaurant in your neighborhood and find out its demand patterns and also identify the underlying reasons.

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9.3 STRATEGIES FOR MATCHING CAPACITY AND DEMAND

Managers can use a wide variety of strategies for matching capacity and demand. This requires a clear understanding of demand patterns as well as the organization's capacity constraints. The strategies to be adopted can be broadly divided into two categories :

- 1) Changing demand to fit supply (capacity) – marketing mix strategies
- 2) Changing supply (capacity) to fit demand – input scheduling strategies

Let us discuss the above two strategies in detail.

1) Strategies for Managing Demand

The organization should determine the optimum level of demand for its given capacity. Once this has been determined, it can vary its marketing mix elements of product, price, place and promotion to change demand in line with the capacity.

- a) **Product:** As a service provider you can alter the service offering to even the demand. The changes in service offering may be seasonal or based on days of the week or time of the day depending on the nature of demand fluctuations. A hotel for example may focus on weekend family entertainment and recreation package to cope up with low demand from business executives during weekends. A management institute may offer more management development programmes during the vacation period of its regular management programme students. However, as marketers you must ensure that by offering different types of services the image or positioning of the service firm is not diluted or confused. Increasing demand during slack period doesn't mean that you should take business from any segment that is available.
- b) **Pricing:** The demand curve suggests that quantity of product demanded varies with the price. Many service marketers reduce price during the periods of low demand to increase the demand. Airlines offer low fares during odd hours like late night flights, movie theaters offer a lower price ticket for the morning show, hotels offer large discounts during off seasons and also higher than normal prices during say Christmas or New year, restaurants and many retail outlets offer happy hours wherein discounts are offered. Using price as an effective tool for managing

demand would require a proper understanding of the demand curve – its shape, slope etc.

Also, you should appreciate the existence of different demand curves for different segments during the same time period. As marketers you may also face an additional challenge when multiple segments are served at the same time and these segments have paid different prices. This requires putting in certain usage conditions (for example for availing low priced fares in airlines customers have to book in advance and there are higher cancellation charges) and/or providing value enhancement to higher paying customers.

- c) **Place (Distribution):** Many service firms modify their time and place of delivery as a strategy to match demand and capacity. Bank may change its timings on specific days or during specific period, finance companies use mobile vans for distribution and collection of forms, hospitals like Apollo have created satellite clinics to deal with routine consultations, tests and medical services.
- d) **Promotion:** You can also shift the demand by properly communicating with your customers. The customers should be made aware of the peak timings of the demand and also the benefits they can get in availing the service during non-peak timings. They should also be properly informed about changes in product, pricing and distribution. This can be done by putting signages at the service outlets (like banks) or advertising. Service firms can also use sales promotion to manage demand. Many airlines offer free ticket for companion in the business class, some business hotels offer free stay for spouse during the weekend stay. Proper promotional strategy can help the service organization in shifting demand from high to low period as well as stimulating demand during low periods.

Activity 3

Select any service organization and analyze how it uses its marketing mix elements to influence demand.

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2) Strategies for Managing Capacity

Managing capacity involves changes in different components of the resources of a firm like people, facilities, equipments, time etc. By making changes in these components you can achieve a better match between demand and capacity.

- a) **Using Part-time Employees:** During periods of peak demand, service firm may hire additional part-time employees. This helps in increasing capacity as well as reducing costs. However, issues like attitude of part-time employees, training concerns, higher turnover etc have to be properly addressed.
- b) **Employees Working Overtime:** Some of the concerns raised above regarding part-time employees can be eliminated by having employees work over time. However, working for longer hours may have adverse impact on service quality and also involves higher costs as overtime charges are generally at higher rates.
- c) **Cross Training Employees:** Cross training of employees results in a flexible capacity, wherein employees can perform several different jobs. Southwest Airlines strongly believes in this philosophy. The same

employee may move from ticketing to gate counters. This helps in avoiding underutilization of resources and also increasing the efficiency of the employees.

- d) **Scheduling:** During peak demand periods people, facilities, and equipments are used at full capacity. However, facilities and equipments also require proper maintenance. This can be scheduled during periods of low demand. Similarly, for human resources, off-peak periods can be used for training purposes as well as for granting vacations.

In addition to the above, service firms can also meaningfully manage its capacity by increasing customer participation (customers can be used as productive resources e.g. self-service in restaurants), outsourcing, modifying the capacity (e.g. reconfiguring hotel rooms), renting facilities or equipments, and taking a subcontract work.

The strategies discussed above for managing demand and capacity have been summarized in Table 9.1

Table 9.1: Strategies for Matching Demand and Capacity

	High Demand/Slack Supply	Slack Demand/Over Supply
Managing Demand (Changing demand to fit capacity)	▫ Educate customers to curtail usage during peak periods (through signages or advertising) ▫ Offer incentives for usage during non-peak periods ▫ Charge full price—no discounts or premium pricing ▫ Take care of regular customers first	▫ Modify service offerings ▫ Offer discounts, sales promotion schemes ▫ Modify hours of operations ▫ Bring service to the customers
Managing Capacity (Changing capacity to match demand)	▫ Hire part-time employees ▫ Keep employees overtime ▫ Cross train employees ▫ Outsource, rent facilities/equipments	▫ Schedule training of employees ▫ Maintenance, repairs, renovations ▫ Schedule employee vacations ▫ Take on subcontract work

9.4 YIELD MANAGEMENT

Yield management is a method for managing demand and capacity profitably by service organizations. It extensively uses computer based technology to study patterns in consumer behaviour in order to manage differential pricing. Yield management has gained widespread acceptance in airline and hotel industry. It helps a firm sell the right capacity to the right type of customer, at the right time for the right price. Put in simple terms,

$$\text{Yield} = \frac{\text{Actual revenue}}{\text{Potential revenue}} = \frac{\text{Actual capacity used} \times \text{Average actual price}}{\text{Total capacity} \times \text{Maximum Price}}$$

It is a measure of the extent to which an organization's resources are being used to their full revenue generation potential. Yield can be increased by an organization by a properly planned differential pricing strategy. Yield management techniques are useful and appropriate: i) when a firm is operating with a relatively fixed capacity, ii) when demand can be segmented, iii) when inventory is perishable, iv) when the product is sold in advance, v) when demand

fluctuates substantially, and vi) where marginal sales cost is low, but capacity change costs are high (Sherly Kimes, 1989). Yield management may require a lot of mathematical programming, economic models and expert systems.

Take the example of an airline. It may not be in a position to sell all its seats at full prices. This may result in a number of seats remaining vacant. Once an aircraft takes off with vacant seats, that capacity is lost for ever. Considering that incremental cost of taking additional passengers is low, airlines offer discounted / apex fares with certain conditions, in order to bring in more revenues and increase the yield. Same is the case with hotels.

$$\text{Yield in case of hotel would be} = \frac{\text{room} - \text{nights sold} \times \text{actual average room rate}}{\text{room} - \text{nights available} \times \text{published room rate}}$$

Take a hypothetical example of a 100 room hotel with a rate of Rs. 2,000 per night i.e. potential revenue is Rs. 2 lakh per night. However, it doesn't get full occupancy at these rates and attract only 50% occupancy, resulting in a yield of 50 percent. On the other hand if the room rates are reduced by 40% i.e. to Rs. 1,200 per night, it attracts 100% occupancy. In this case the yield

becomes 60%

However, a combination of the two can be a better alternative for the hotel. Selling 50% of the rooms at full tariff and remaining 50% at reduced tariff of

Rs. 1,200 would result in higher yield $\left(\frac{160,000}{200,000} \right)$ of 80 percent.

In order to make yield management successful, the marketers should identify the main market segments being served / those can be served. The next step would be to do a proper sales forecast for each segment at particular price levels. Based on this forecast a proper mix of different customer segments at different times can be suggested for maximizing the yield. These steps should be done on a continuous basis so as to adjust to the changing market conditions as well as making use of greater information available about segment wise demand patterns.

Managerial Implications

Sherly Kimes in her widely quoted article "Yield management – A tool for capacity constrained service firms (Journal of Operations Management, Oct. 1989) has identified a number of management issues to be taken into account while implementing a yield management system. While yield management may give a firm competitive advantage, it could also result in the following :

- **Loss of competitive focus :** As most yield management systems focus on maximizing revenue or yield, it may result in neglecting important issues like service quality. That is to say, short term profit maximization may overshadow long term competitive advantages.
- **Customer Alienation :** A customer who pays a higher price for a service than other customers may feel alienated and dissatisfied. He may consider it to be unfair to him. Therefore proper customer education should be an integral part of any yield management system to be effective.
- **Employee Morale Problem:** Yield Management seems to take away discretion from sales and reservation people. Therefore, it should be properly structured to allow for some judgment on the part of the employees.
- **Lack of Employee Training :** A yield management system will require extensive training of all employees. They should understand its objectives as well as its operation and how it affects their jobs.

9.5 MANAGING CUSTOMER WAITING

In the previous sections you have learnt about demand patterns and strategies to match demand and capacity. However sometimes it is not possible to match demand and capacity and waiting by customers become inevitable. Waiting is a common phenomenon at hospitals, restaurants, banks, hair cutting saloons etc. In such situations waiting time becomes one of the key factors in consumer's evaluation of service. While reducing waiting time is important for marketers, it is equally if not more, important to reduce the customer's perceived waiting time. If a customer's perceived waiting time is less, he will be more satisfied with the service. Thus service waits can be controlled by two broad techniques viz. Operations Management and Perception Management.

- 1) **Operations Management** : It involves reducing the amount of time customers have to wait. This can be done in a number of ways. Firstly the firm should analyze its operational processes in order to identify and remove inefficiencies or bottlenecks, if any. Secondly, in case waiting cannot be avoided, a reservation system can be used. This will help in getting the customer out of a queue. Thirdly, customers can be encouraged to use the facilities during non-peak hours. Fourthly, greater use of information technology can be made wherein customers can use telephone, computers, etc to conduct business. For example, banks can deploy ATMs, provide phone banking and internet banking to reduce pressure at the branches. Lastly, as marketers you can also differentiate waiting customers wherein some customers may wait for more time while others receive a quicker service. The differentiation can be done on the basis of a number of factors like importance of the customer, urgency of the job, duration of the service transaction and payment of a premium price. In case queues cannot be avoided, the organization has to decide on the type of queuing system to be adopted. There are number of possibilities in this regard. Take for example the computerized railway reservation centres wherein there are multiple-queues and the customer has the option to join whichever queue he wants to and can also switch over to other queue if the wait appears to be shorter in that. Another option is to have a single queue system wherein first cum first serve rule applies to everyone. A slight variation of single queue system can be that each customer on arrival is given a number and waits at the reception area enabling the customer to sit, relax and mix up with other customers.

Exhibit 9.1. Managing Waiting Lines at Sri Venkateswara Temple, Tirupati

More than fifty thousand pilgrims visit the Sri Venkateswara Temple every day. 'Sarvadarsanam' (darshan for all) timings are different on different days of week. On normal days, about 18 hours are allotted for Sarvadarsanam and on peak days it is open for 20 hours. There is also a provision of 'Special Darshan' on purchase of tickets. Pilgrims who use the queue for Special Darshan have a shorter waiting time. This queue merges with the Sarvadarsanam queue at a fixed point and the darshan timings are the same as that for Sarvadarsanam.

The Sudarsanam token system was introduced to minimize the waiting time for Sarvadarsanam, Special Darshan and other paid darshan/sevas. Some of its features:

- The tokens are available free of cost at a number of convenient places in the town.
- The time for darshan is indicated on the token.
- Pilgrims can enter the Vaikuntam Queue Complex at Tirumala at the time indicated on

the tokens.

- They can have darshan within two hours of entering the Queue Complex.
- As this system saves on waiting time, it provides pilgrims with enough time to visit temples in the vicinity.
- In order to keep a track of the number of pilgrims and ensure their smooth flow, one token is issued per head. Collective tokens for groups are not issued.

Source: www.tirumala.org

2) **Perception Management:** Limited success of operations management in waiting line management has led to increased interest in managing the perceptions of wait experience. If you cannot control the actual wait duration, then control the customer's perception of it. Maister has proposed following eight principles that you can use as service marketers to influence customer's perception of waits and their satisfaction with waiting lines

- Unoccupied time feels longer than occupied time.
- Preprocess waits feel longer than in-process waits.
- Anxiety makes waits seem longer.
- Uncertain waits are longer than known, finite waits.
- Unexplained waits are longer than explained waits.
- Unfair waits are longer than equitable waits
- The more valuable the service, the longer the people will wait.
- Solo waiting feels longer than group waiting.

Therefore, you should appreciate that though operations management techniques are important, however, while developing strategies for waiting lines you should never overlook the effects of perceptions management. The following suggestions can be used in order to make waiting fun or at least tolerable.

- 1) Determine the acceptable waiting time for your customers.
- 2) Since unoccupied time feels longer than occupied time, keep customers occupied by installing distractions that entertain and physically involve them. For example, television sets can be installed in the waiting areas, magazines or reading materials related to the service can be provided. Exhibit 9.2 provides an interesting illustration in this regard.
- 3) Provide 'waiting duration information' i.e. information about the expected length of a wait and/or 'queuing information' i.e. a consumer's position in the queue, with continuous updates. Michael Hue & David Tse suggest that in short waits, no information is needed. In case of intermediate waits, waiting duration information appears to be a better choice than queuing information. However, in case of long waits, waiting duration information may be less effective than queuing information. Also providing queuing information is more important as compared to waiting duration information when service organization has difficulty in accurately estimating the length of wait or when the waiting line is not visible to customers.
- 4) If unexpected delays occur, explanation should be given to the customers. This helps in reducing uncertainty and customer irritation. The key is to impress upon the customer that he has not been forgotten. Simple things like providing a glass of water or a cup of tea to the waiting customer can help.
- 5) Try to modify customer arrival behaviour.
- 6) Keep resources not serving customers out of sight. This can be done by keeping idle employees out of view and conducting activities that do not involve customer interactions out of the customer's sight.
- 7) Try to reduce pre-service waiting by transferring some of the pre-service waiting to the service encounter phase. For example, menu cards may be

provided to the customers while waiting, to decide on their orders, medical information may be collected from the patient prior to actually meeting the doctor.

- 8) A smiling service person who knows his job well can be very helpful in overcoming many negative effects of waiting. Therefore, training and incentives / rewards for providing good service should be made.

Exhibit 9.2. "Waiting in line : Experiment by Bank of America"

Bank of America through their review of data realized that there might be opportunities to reduce perceived wait times without reducing actual wait times. An earlier study by market researchers had also revealed (the study was conducted by intercepting some 1000 customers standing in bank lines) that after a person stands in line for more than three minutes, a wide gap opens between actual and perceived times. While a two-minute wait may usually feel like a two-minute wait, a five minute wait may feel like a ten-minute wait. Also, psychological studies have revealed that if you distract a person from a boring chore, time seems to pass much faster.

In the summer of 2001 the bank installed television sets over the teller booths at one of its branches to test its hypothesis that "if you entertain people in line by putting television sets in the transaction zone – above the row of tellers in a branch lobby – you will reduce perceived wait times by at least 15%". The results obtained were significant. After the installation of the T.V. sets the degree of overestimation of wait times dropped from 32% to 15% at the test branch. Before the implementation of this experiment, customers who waited longer than five minutes, significantly overestimated their waiting time by 32%. (Average actual time : 6.17 minute, Average perceived time : 8.16 min). After the installation of monitors in the bank lobby these overestimates for the same customer groups dropped to 15% (Average actual time : 6.14 minutes, Average perceived time : 7.04 min). Considering that long waits have direct impact on customer satisfaction, the bank through a research also concluded that a branch with more than a thousand households in its customer base would be able to recoup up the cost of installing T.V. sets in less than a year because of increased customer purchases and retention due to higher customer satisfaction. (Based on the study analysis that every one point improvement in bank's customer satisfaction index added \$1.40 in annual revenue per household and that the reduction in perceived wait times would translate into a 5.9 point increase in overall banking-centre customer satisfaction)

Source: Stefan Thomke, "R&D Comes to Service: Bank of America's Path breaking Experiments," Harvard Business Review, April 2003, p.76-77.

9.6 MANAGING DEMAND AND WAITING LINES: CASE OF AN AMUSEMENT PARK

Introduction: In this section we are giving you a brief case situation concerning an amusement park. Also given are the comments and possible solutions to the problems raised, given by top executives of service companies (The case situation and the comments / solutions have been excerpted from a Harvard Business Review Case Study – details given in sources at the bottom of this section). In the last part of this section we have also brought out some of the innovative practices adopted by successful amusement parks with regards to managing demand and waiting lines. Before we move over to the case situation, let us first take a brief look at the amusement park industry as such.

The industry has its origin about 400 years ago in the Danish capital Copenhagen. In India the industry is in the growth stage with around 1000 crore said to have been invested in the last few years. This growth is primarily a result of higher disposable incomes and an increasing willingness on the part of the customers to consider new forms of entertainment. Foreign companies like Universal Studio, Time Warner and Disney are eyeing the Indian market. Presently the leading players in the Indian market include Appu Ghar in Delhi and Essel World near Mumbai with a lot of new players like Sammy's Dreamland in Bangalore also coming up.

Case Situation: The amusement park with a successful history was now facing problems. It had its first money losing year last year followed by another one now. The park had three ways to bring in more revenues: increase visit per customer, increase average spending per visit or attract new customers. Because of a mature industry (U.S.A) all three were hard to do. As pulling in people from broader geographical areas seemed an unlikely proposition due to the wide availability of such parks, attracting new customers required new value proposition. With this background, it was proposed to offer a “preferred guest card” to win more business from moneyed and time pressed group of people. Under this plan, visitors could pay an additional fee to get free rein of the park: Card Holders would enter the ride through separate lines which would give them first crack and they would be seated immediately at any in-park restaurant. It was hoped that this plan will help to up-sell the people who are already coming to the park. And by making it possible to spend less time in queues, the guest card will also attract a different type of customer – time starved, high-income professionals and their families, who might otherwise avoid the whole experience. However, certain objections were raised against such a scheme. “I don’t even think it’s a great experience for the preferred guests. Who want to feel all that animosity diverted at them? The key to this business is the customers feeling good while they are here. With this scheme neither side’s coming back” commented an executive. A possible solution given to this was to separate the lines and limiting the percentage of special tickets issued on any given day. If the ‘preferred guest card’ scheme was not implemented the park might be forced to raise price across the board.

Before moving on to the next part, analyze the above situation and identify possible solutions.

What Experts Say : In this part comments of three senior executives from different service industries have been briefly given

- i) A cofounder of a premium health care service:
 - Creating two types of service at different prices will create problem for the park.
 - Raise the admission price instead
- ii) Chief Marketing office of a leading bank:
 - It makes good business sense to segment customers and to offer a different level of service-at a higher price-to those at the upper end of the market.
 - The key is to do it discreetly and in a way that does not degrade the quality of service to the basic customer.
 - The expedited line should be hidden from the view of those waiting in longer one.
- iii) A former CEO of large Airlines:
 - Service differentiation at an amusement park must be subtler than what has been proposed.
 - The park can offer all its guests the opportunity to reserve a time slot for a particular ride. A fixed number of seats may be allocated to reservation (say a third or a half) and then give the customers lining up an option to wait or to make ride reservation for later in the day.

Some Practices in Successful Amusements Parks: Disney lands offer a form of reservation – Fastpass – by which guests may go up to one of the rides offering the Fastpass service and obtain reservation to come at a certain

At the specified time they come back and bypass the waiting line saving an hour or more in waiting. However, the customers are limited to one Fastpass every four hours in order to ensure that the rides are able to accommodate both Fastpass and regular customers. Further, line management at Disney also involves continuously entertaining waiting customers and providing them with the information regarding the duration of their waits. Signs are posted at intervals in the queue, stating the expected time until service commencement. Because of Fastpass, rather than waiting in line, customers spend more money in the restaurant and shops. Dreamworld in Australia caters to both international as well as local customers. Considering different price sensitivities of local and international customers (while a Japanese couple will pay \$78 - \$39 per person entrance fees – without hesitation as this is a small portion of their vacations, a local family of four looking for weekend entertainment may view \$136- \$39 adult charge and \$ 29 charge for children — amounts to a large portion of their entertainment expenses). Dreamworld offers a separate package to local customers involving yearly passes at much lower prices. It is a common practice for tourist attractions such as Dreamworld to offer special rates for local residents. It is however very important for the parks to know its customer mix.

Sources

- i) Economic Times, New Delhi – Dec 16, 2003 – “Foreign majors eye amusement park industry” by Rahul Sachitanand.
- ii) Harvard Business Review, Nov. 2001, HBR Case Study “ Are Some Customers More Equal Than Others ?” by Nunes and Johnson, pp 37-50.
- iii) “Marketing for Hospitality and Tourism”, 3rd edn., by Kotler, Bowen and Makens, Prentice Hall, 2003 – page 418, 459-460.
- iv) “Waiting for Services : The Relationship between Delays and Evaluations of Service,” by Shirley Taylor, Journal of Marketing, April 1994, p.66.

Activity 4

Visit any amusement park and/or talk to your friends about their experience at amusement parks. Identify strategies adopted by the parks to manage demand (e.g. on weekdays vs. weekends, special package for specific customers groups etc.) and waiting lines.

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9.7 SUMMARY

This unit deals with issues related to managing demand and capacity in service organizations. Considering the fact that services are perishable, demand fluctuation is considered to be a somewhat serious problem for services marketers. A service organization with a fixed capacity may be faced with four different situations viz. excess demand, demand exceeding the optimum capacity level, demand and supply being balanced at the optimum capacity and excess capacity. The first step towards developing strategies for matching demand and capacity is to study the demand patterns and the underlying causes. The strategies for matching demand and capacity can be broadly divided into two categories – changing demand to fit supply and changing supply to fit demand. These two strategies have been discussed in detail in this unit. Subsequently yield management technique for managing demand and

capacity profitably has been explained. Despite strategies for matching demand and capacity being in place, waiting by customers becomes inevitable in a number of service industries. Service waits can be controlled by operations management and perceptions management. This unit outlines certain suggestions which can help you in better management of waiting lines. The last section of the unit gives you a brief case situation concerning demand management and waiting lines issues in an amusement park.

9.8 SELF ASSESSMENT QUESTIONS

A) Objective Types Questions

1. Which of the following strategies would you recommend to manage demand when it is too high?
 - a. offer discounts
 - b. bring the service to the customers
 - c. offer incentive to customers for usage during non-peak times
 - d. all of the above
2. Providing separate check-in lines for first class passengers by an airlines is an example of differentiating waiting customers on the basis of
 - a. urgency of the job
 - b. payment of premium price
 - c. duration of service transaction
 - d. none of the above
3. Which of the following is not true regarding waiting by customers?
 - a. uncertain waits are longer than known waits
 - b. unexplained waits are longer than explained waits
 - c. in-process waits feels longer than the pre-process waits
 - d. unoccupied time feels longer than occupied time.
4. In which of the following demand conditions would you witness the situation in which no customer is being turned away but the quality of service may still suffer due to crowding or staffing being pushed beyond their abilities to deliver consistent quality?
 - a. excess demand
 - b. demand exceeds optimum capacity
 - c. demand and supply are balanced at the level of optimum capacity
 - d. excess capacity
5. Appropriate situation for effective yield management application includes:
 - a. ability to segment markets
 - b. product sold in advance
 - c. fluctuating demand
 - d. all of the above
6. Which of the following strategies for flexing capacity to match demand would be appropriate when demand is too low?
 - a. perform maintenance, repairs
 - b. schedule vacations
 - c. schedule employee training
 - d. all of the above

Strategy 7. Several major restaurant chains offer discounts on days when business is normally slow. This strategy is employed because service are

Managing Demand and Capacity

- a. perishable
- b. variable
- c. inseparable
- d. intangible

Answers

1. c 2. b 3. c 4. b 5. d
6. d 7. a

B. Discussion Questions

1. Explain why is it important for service organizations to match demand and capacity. What are the implications of a mismatch between the two?
2. Explain the significance of determining the demand patterns. Select any service organization of your choice and describe its demand patterns and its underlying causes.
3. Describe the strategies for matching supply and demand giving suitable examples.
4. Explain the term 'Yield Management'. Identify some of the managerial issues to be taken into account while implementing a yield management system.
5. Select a service organization you are familiar with, where customers have to wait in line for service. Develop a waiting line strategy for the organization.

9.9 FURTHER READINGS

- Donald J Shemwell, Jr. And J. Joseph Cronin, Jr. "Services Marketing Strategies for coping with demand/supply imbalances" Journal for Services Marketing, Vol. 8, No. 4, 1994.
- Shirley Taylor, "Waiting for Service: The relationship between delays and evaluation of services" Journal of Marketing, April 1994.
- Sherlyl E. Kimes, "Yield Management: A Tool for Capacity-Constrained Service Firms" Journal of Operations Management, Vol. 8 No. 4, 1989.
- Karen Katz, Blaire Larson, Richard Larson, "Perceptions for the waiting-in line blues: Entertain, Enlighten and Engage" Sloan Management Review, Winter 1991.
- Paul F. Nunes and Brain A. Johnson, HBR Case Study "Are Some Customers More Equal Than Others", Harvard Business Review, Nov. 2001.
- Stefen Thomke, "R & D Comes to Services: Bank of America's Pathbreaking Experiments", Harvard Business Review, April 2003.
- Valarie Zeithaml, A. Parasuraman & Leonard Berry "Problems and Strategies in Services Marketing", Journal of Marketing, Spring 1985.
- Micheal Hui and David Tse, "What to tell customer in waits of Different Lengths: An Integrative Model of Service Evaluation", Journal of Marketing, Vol. 60, April 1996.

UNIT 8 INVENTORY MODELS

Structure

- 8.1 Introduction
 - Objectives
- 8.2 Inventory Control
- 8.3 Economic Order Quantity (EOQ) Model with Uniform Demand
- 8.4 EOQ Model with Different Rates of Demand in Different Cycles
- 8.5 EOQ Model when Shortages are Allowed
- 8.6 EOQ Model with Uniform Replenishment
- 8.7 EOQ Model with Price (or Quantity) Discounts
- 8.8 Summary
- 8.9 Solutions/Answers

8.1 INTRODUCTION

In Unit 7, you have studied the sequencing problem, which involves determining the optimum order or sequence of jobs for a process to optimise the total time. We have discussed two types of sequencing problems: the ones with n number of jobs to be completed through 2 machines and those with 2 jobs to be completed through m number of machines, in some pre-assigned order.

In this unit, we discuss various inventory models. Inventory refers to a stock of goods, materials, human resources or financial resources or any other idle resource having some economic value, which is stocked in order to meet the demand expected in future. Almost every business must maintain an inventory for running its operations efficiently and smoothly.

Although inventories are essential for business, maintenance of inventories also costs money by way of expenses on stores, equipment, personnel, insurance, etc. Thus, excess inventories are undesirable. This calls for controlling the inventories in the most profitable way. In the present unit, we discuss the models for inventory control known as **economic order quantity** models. These models help in deciding as to how much quantity should be kept in stock in order to balance the costs of holding too much stock vis-à-vis the costs of ordering in small quantities.

In this unit, we discuss inventory control and various factors involved in inventory analysis in Sec. 8.2. In Secs. 8.3 to 8.7, we describe five models for determining the economic order quantity: i) when demand is uniform; ii) when rates of demand are different in different cycles; iii) when shortages are allowed; iv) when replenishment is uniform; and v) when price (or quantity) discounts are given.

Objectives

After studying this unit, you should be able to:

- explain the concept of inventory control;
- determine the economic order quantity when demand is uniform;
- determine the economic order quantity when rates of demand are different in different cycles;

- determine the economic order quantity when shortages are allowed;
- determine the economic order quantity when replenishment is uniform; and
- determine the economic order quantity when there are price discounts.

8.2 INVENTORY CONTROL

An inventory means a physical stock of idle resources of any kind having some economic value kept for the purpose of meeting future demand. It indicates the raw material required before production, the finished goods after production ready for delivery to consumers, human resources, financial resources, etc., which are stocked in order to meet an expected demand in the future. Almost every business must maintain an inventory for running its operations efficiently and smoothly. If an enterprise does not maintain an inventory, it may suddenly find at some point in its operations that it has no materials or goods to supply to its customers. Then on receiving a sales order, it will first have to place order for purchase of raw materials, wait for their receipt and then start production. The customer will, thus, have to wait for a long time for the delivery of the goods and may turn to other suppliers, resulting in loss of business/goodwill for the enterprise.

Maintaining an inventory is necessary because of the following reasons:

- It helps in smooth and efficient running of an enterprise.
- It provides service to the customer at short notice. Timely delivery can fetch more goodwill and orders.
- In the absence of the inventory, an enterprise may have to pay high prices because of piecemeal purchasing. Maintaining an inventory may earn price discounts because of bulk purchasing. Such purchases entail less orders and, therefore, less clerical costs.
- It also takes advantage of favourable market.
- It acts as a buffer stock when raw materials are received late and shop rejections are too many.
- Process and movement inventories (also called pipeline stocks) are quite necessary in big enterprises wherein a significant amount of time is required to ship items from one location to another.

Though inventories are essential, their maintenance also costs money by way of expenses on stores, equipment, personnel, insurance, etc. Thus, excess inventories are undesirable. So, only that quantity should be kept in stock, which balances the costs of holding too much stock vis-à-vis the costs of ordering in small quantities. This calls for controlling the inventories in the most profitable way and that is why we need inventory analysis. We now discuss various factors involved in inventory analysis.

1. Inventory related costs

Various costs associated with inventory control are often classified as follows:

- Set-up cost:** This is the cost associated with the setting up of machinery before starting production. The set-up cost is generally assumed to be independent of the quantity ordered for.
- Ordering cost:** This is the cost incurred each time an order is placed. This cost includes the administrative costs (paper work, telephone calls, postage), transportation, receiving and inspection of goods, etc.

- iii) **Purchase (or production) cost:** It is the actual price at which an item is purchased (or produced). It may be constant or variable. It becomes variable when quantity discounts are allowed for purchases above a certain quantity.
- iv) **Carrying (or holding) cost:** The cost includes the following costs for maintaining the inventory: i) Rent for the space; ii) cost of equipment or any other special arrangement for storage; iii) interest of the money blocked; iv) the expenses on stationery; v) wages of the staff required for the purpose; vi) insurance and depreciation; and vii) deterioration and obsolescence, etc.
- v) **Shortage (or Stock-out) cost:** This is the penalty cost for running out of stock, i.e., when an item cannot be supplied on the customer's demand. These costs include the loss of potential profit through sales of items demanded and loss of goodwill in terms of permanent loss of the customer.

2. Demand

Demand is the number of units required per period and may either be known exactly or known in terms of probabilities. Problems in which demand is known and fixed are called **deterministic problems** whereas problems in which demand is known in terms of probabilities are called **probabilistic problems**.

3. Selling Price

The amount which one gets on selling an item is called its selling price. The unit selling price may be constant or variable, depending upon whether quantity discount is allowed or not.

4. Order Cycle

The period between placement of two successive orders is referred to as an order cycle. The order may be placed on the basis of either of the following two types of inventory review systems:

- a) The record of the inventory level is checked continuously until a specified point is reached where a new order is placed. This is called continuous review.
- b) The inventory levels are reviewed at equal intervals of time and orders are placed accordingly at such levels. This is called periodic review.

5. Time Horizon

The period over which the time cost will be minimised and inventory level will be controlled is termed as time horizon. This can be finite or infinite depending on the nature of demand.

6. Stock Replenishment

The rate at which items are added to the inventory is called the rate of replenishment. The actual replenishment of items may occur at a uniform rate or be instantaneous over time. Usually uniform replacement occurs in cases when the item is manufactured within the factory while instantaneous replacement occurs in cases when the items are purchased from outside sources.

7. Lead Time

The time gap between placing an order for an item and actually receiving the item into the inventory is referred to as lead time.

8. Reorder Level

The lower limit for the stock is fixed at which the purchasing activities must be started for replenishment. With this replenishment, the stock reached at a level is known as maximum stock. The level between maximum and minimum stock is known as the reorder level.

9. Economic Order Quantity (EOQ)

The order in quantity that balances the costs of holding too much stock vis-à-vis the costs of ordering in small quantities too frequently is called Economic Order Quantity (or Economic lot size).

10. Reorder Quantity

The quantity ordered at the level of minimum stock is known as the reorder quantity. In certain cases it is the 'Economic Order Quantity'.

In Secs. 8.3 to 8.7, we shall discuss the following inventory models for obtaining economic order quantity:

- i) EOQ Model with Uniform Demand
- ii) EOQ Model with Different Rates of Demand in Different Cycles
- iii) EOQ Model when Shortages are Allowed
- iv) EOQ Model with Uniform Replenishment
- v) EOQ Model with Price (or Quantity) Discounts

However, before discussing these models, we give the notations that we shall use in the development of the models.

The notation used in the Models

Q = Number of units ordered (supplied) per order

D = Demand in units of inventory per year

N = Number of orders placed per year

TC = Total Inventory cost

C_o = Ordering cost per order

C = Purchase or manufacturing price per unit inventory

C_h = Carrying or holding cost per unit per period of time the inventory is kept

C_s = Shortage cost per unit of inventory

t = The elapsed time between placement of two successive orders

r_p = Replenishment rate at which lot size Q is added to inventory.

8.3 ECONOMIC ORDER QUANTITY (EOQ) MODEL WITH UNIFORM DEMAND

The objective of the **EOQ model with uniform demand** is to determine an optimum economic order quantity such that the total inventory cost is minimised. We make the following assumptions for this model:

1. Demand rate (D) is constant and known;
2. Replenishment rate (r_p) is instantaneous;
3. Lead time is constant and zero;

4. Purchase price is constant, i.e., discounts are not allowed;
5. Carrying cost and ordering cost are known and constant; and
6. Shortages are not allowed.

The situation can be graphically represented as shown in Fig. 8.1.

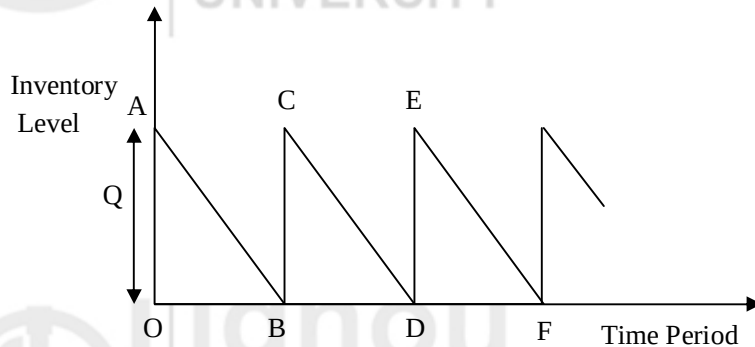


Fig. 8.1

The graph in Fig. 8.1 shows that initially there were Q units in the stock. The number of units goes on decreasing with respect to time to meet the demand and this is represented by the line AB in the graph. When the stock vanishes, i.e., the point B is reached, the stock level rises to Q instantaneously as the replenishment is instantaneous. Since the demand is uniform, the rate of decrease of the quantity remains the same as earlier. Therefore, this is represented by the line CD in the graph, which is parallel to AB . Similarly, EF is parallel to AB and CD due to uniform demand, and so on.

Since the demand is uniform, the average inventory is simply the arithmetic mean of the maximum and the minimum levels of inventory. Let Q be the quantity ordered (or replenished) when the minimum level, i.e., zero is reached.

Therefore, the average inventory $= \frac{Q+0}{2} = \frac{Q}{2}$

The ordering cost = Number of orders per year $\times$ ordering cost per order

$$= N \times C_o = \frac{D}{Q} \times C_o$$

The carrying cost = Average units in inventory $\times$ carrying cost per unit

$$= \frac{Q}{2} \times C_h$$

The total inventory cost is the ordering cost plus the carrying cost. Therefore,

$$TC = \frac{D}{Q} C_o + \frac{Q}{2} C_h$$

The total inventory cost (TC) is minimum at that value of Q where the derivative of TC with respect to Q is zero. Differentiating TC with respect to Q and then equating it to zero, we get

$$-\frac{D}{Q^2} C_o + \frac{1}{2} C_h = 0$$

$$\Rightarrow \frac{D}{Q} C_o = \frac{Q}{2} C_h \quad (\text{i.e., ordering cost} = \text{carrying cost})$$

$$\Rightarrow Q^2 = \frac{2D C_o}{C_h} \Rightarrow Q = \sqrt{\frac{2D C_o}{C_h}}$$

This value of Q minimises the total inventory cost (TC) and hence it is the economic order quantity. Let us denote it by Q^* . Thus, the EOQ for this model is

$$Q^* = \sqrt{\frac{2D C_o}{C_h}}$$

Hence, the optimum number of orders placed per year (N^*) = $\frac{D}{Q^*}$

The minimum total yearly inventory cost

$$\begin{aligned} (TC^*) &= \frac{D}{Q^*} C_o + \frac{Q^*}{2} C_h = \frac{2D C_o + (Q^*)^2 C_h}{2Q^*} \\ &= \sqrt{2D C_o C_h} \quad (\text{on simplification}) \end{aligned}$$

and the total minimum cost = (TC^*) + cost of material.

Let us consider an example to obtain the total minimum cost in this model.

Example 1: An enterprise requires 1000 units per month. The ordering cost is estimated to be ₹ 50 per order. In addition to ₹ 1, the carrying costs are 5% per unit of average inventory per year. The purchase price is ₹ 20 per unit. Find the economic lot size to be ordered and the total minimum cost.

Solution: We are given that

$$\begin{aligned} D &= \text{Monthly demand} \times 12 \\ &= 1000 \times 12 = 12000 \text{ units per year} \end{aligned}$$

$$C_o = ₹ 50 \text{ per order}; C = ₹ 20 \text{ per unit}$$

$$C_h = 1 + 5\% \text{ of } ₹ 20$$

$$= 1 + 1 = ₹ 2 \text{ per unit of average inventory}$$

The economic lot size, therefore, is given by

$$Q^* = \sqrt{\frac{2D C_o}{C_h}} = \sqrt{\frac{2 \times 12000 \times 50}{2}} = 775 \text{ units}$$

Total minimum cost = TC^* + Cost of material

$$\begin{aligned} &= \sqrt{2D C_o C_h} + (12000 \times 20) \\ &= \sqrt{2 \times 12000 \times 50 \times 2} + 240000 \\ &= \sqrt{2400000} + 240000 = \sqrt{240} \times 100 + 240000 \\ &= 15.5 \times 100 + 240000 = ₹ 241550 \end{aligned}$$

You may now like to solve the following problems to assess your understanding.

- E1)** The XYZ manufacturing company uses 12000 units of raw material annually, which costs ₹1.25 per unit. Placing each order costs ₹15 and the carrying costs are 15% per year per unit of the average inventory. Find the economic order quantity.
- E2)** The XYZ manufacturing company has determined from an analysis of its accounting and production data for a part, that its cost to purchase is ₹36 per order and ₹2 per part. Its inventory carrying charge is 18% of the average inventory. The demand of this part is 10000 units per annum.
- What should the economic order quantity be?
 - What is the optimum number of days supply per optimum order?

8.4 EOQ MODEL WITH DIFFERENT RATES OF DEMANDS IN DIFFERENT CYCLES

In this EOQ model, the same quantity is ordered each time when the stock vanishes with the assumption that the replenishment is instantaneous. The demand is not uniform and the stock vanishes in different periods of time. The situation can be graphically represented as shown in Fig. 8.2.

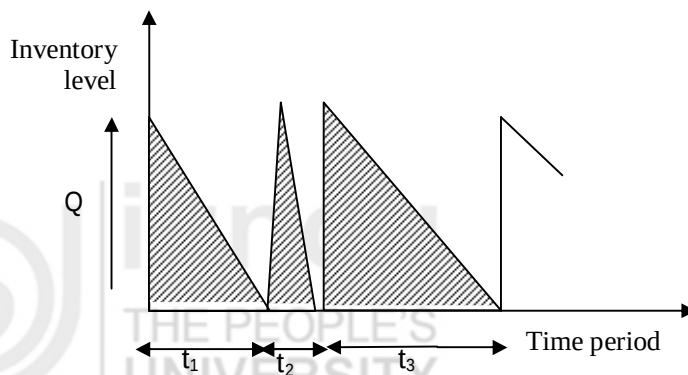


Fig. 8.2

Let the demand in different periods of time $t_1, t_2, \dots, t_n$ be $D_1, D_2, \dots, D_n$, respectively, so that the total demand in time T is given by

$$D = D_1 + D_2 + \dots + D_n \quad \text{where } T = t_1 + t_2 + \dots + t_n$$

The cost of ordering in time T

$$\begin{aligned}
 &= \text{No. of orders per year} \times \text{ordering cost per order} \\
 &= NC_o \\
 &= \frac{\text{Total Demand in time } T \times \text{Ordering cost per order}}{\text{No. of units ordered per order}} \\
 &= \frac{D}{Q} \times C_o
 \end{aligned}$$

The carrying cost for time T

$$= \text{Average unit in inventory} \times \text{carrying cost per unit}$$

$$= \frac{Q \times t_1}{2} C_h + \frac{Q \times t_2}{2} C_h + \dots + \frac{Q \times t_n}{2} C_h$$

$$= \frac{Q}{2} C_h [t_1 + t_2 + \dots + t_n] = \frac{QC_h T}{2}$$

∴ The total inventory cost (TC) = ordering cost + carrying cost

$$TC = \frac{DC_o}{Q} + \frac{QC_h T}{2}$$

This cost is minimum when its derivative with respect to Q is zero, i.e., if

$$-\frac{D}{Q^2} C_o + \frac{1}{2} TC_h = 0$$

$$\Rightarrow \frac{D}{Q} C_o = \frac{Q}{2} TC_h \quad (\text{i.e., ordering cost} = \text{carrying cost})$$

$$\Rightarrow Q^2 = \frac{2D C_o}{C_h T} \quad \text{p} \quad Q^* = \sqrt{\frac{2D C_o}{C_h T}}$$

This result is similar to that of the model discussed in Sec. 8.3. The only difference is that the uniform demand (D) is replaced by the average demand (D/T).

$$(TC^*) = \frac{DC_o}{Q^*} + \frac{Q^* TC_h}{2} = \frac{2DC_o + (Q^*)^2 TC_h}{2Q^*}$$

$$= \sqrt{2DC_o C_h T} \quad (\text{on simplification});$$

The total minimum cost = (TC*) + Cost of material

If T = 1 year, the result of this model is exactly the same as the EOQ model with uniform demand.

8.5 EOQ MODEL WHEN SHORTAGES ARE ALLOWED

The assumptions made for this model are the same as those for the EOQ model with uniform demand discussed in Sec. 8.3, except that shortages are allowed, which may occur frequently. The cost of shortages is assumed to be directly proportional to the number of units short. The assumptions for the model are:

- i) Demand is constant and known;
- ii) Replenishment is instantaneous;
- iii) Lead time is zero;
- iv) Purchasing cost is constant and known;
- v) Carrying cost and ordering cost are constant and known; and
- vi) Shortage is allowed and the shortage cost is directly proportional to the number of units short.

The situation can be represented graphically as shown in Fig. 8.3.

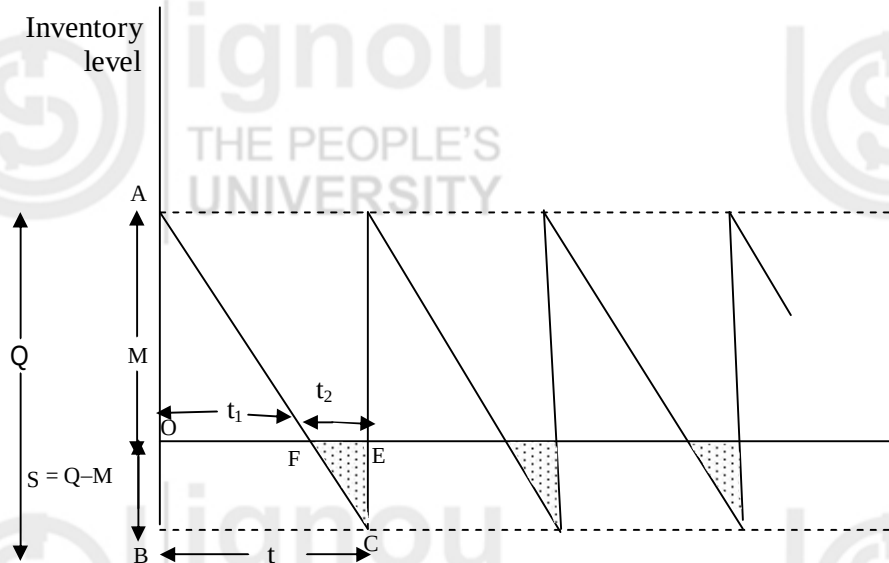


Fig. 8.3

Here,

Q = No. of units ordered per order

D = No. of units required per year

t_1 = Time when there are no shortages

t_2 = Time period during which there is shortage

t = Total cycle time, i.e., $t = t_1 + t_2$

S = Maximum shortage

The same quantity Q is ordered and received every time. The maximum shortage is equal to S . The remaining quantity M is placed in the inventory as surplus to satisfy the demand in the next cycle. Note that S units out of Q are always in the shortage list, i.e., these are never placed in the inventory. Thus, it results in a saving on the inventory carrying cost.

$$\text{No. of orders placed per year} = N = \frac{D}{Q}$$

Ordering cost = No. of orders placed per year $\times$ ordering cost per order

$$= NC_o = \frac{D}{Q} C_o$$

$$\text{Average inventory} = \frac{M}{2}$$

$$\text{Carrying cost per cycle} = C_h \left(\frac{M}{2} \times t_1 \right)$$

Note from Fig. 8.3 that $\triangle ABC$ is similar to $\triangle AOF$. Therefore,

$$\frac{t_1}{t} = \frac{M}{Q} \Rightarrow t_1 = \frac{M}{Q} t$$

$$\text{Therefore, carrying cost} = \left(\frac{M^2 \times t}{2Q} \right) C_h$$

$$\text{Carrying cost per unit time} = \frac{M^2}{2Q} C_h$$

Shortage cost = Average no. of units short $\times$ time period of shortage
 $\times$ shortage cost per unit per time period

$$= \frac{S}{2} \times t_2 \times C_s = \left(\frac{S}{2} t_2 \right) C_s$$

Now, $\triangle ABC$ is similar to $\triangle CEF$ in Fig. 8.3. Therefore,

$$\frac{t_2}{t} = \frac{S}{Q} \Rightarrow t_2 = \frac{S}{Q} t$$

$$\text{Hence, shortage cost} = \left(\frac{S^2 \times t}{2Q} \right) C_s$$

$$\text{Shortage cost per unit time} = \frac{S^2}{2Q} C_s$$

Therefore, the total inventory cost is given as:

$$TC = \text{ordering cost} + \text{carrying cost} + \text{shortage cost}$$

$$\text{or } TC = \frac{D}{Q} C_o + \frac{M^2}{2Q} C_h + \frac{S^2}{2Q} C_s$$

$$\text{or } TC = \frac{D}{Q} C_o + \frac{M^2 C_h}{2Q} + \frac{(Q - M)^2 C_s}{2Q} \quad \dots(1)$$

Now, TC is a function of two variables Q and M. Therefore, to find the minimum inventory cost, we differentiate TC with respect to Q and M separately and equate them to zero. Thus, we get

$$0 = \frac{d}{dQ}(TC) = -\frac{DC_o}{Q^2} - \frac{M^2 C_h}{2Q^2} + \frac{1}{2} \left[\frac{2Q(Q - M)^2 - (Q - M)^2}{Q^2} \right] C_s$$

$$0 = -\frac{DC_o}{Q^2} - \frac{M^2 C_h}{2Q^2} + \left(\frac{Q^2 - M^2}{2Q^2} \right) C_s$$

$$\Rightarrow -\frac{DC_o}{Q^2} - \frac{M^2}{2Q^2} (C_h + C_s) + \frac{C_s}{2} = 0$$

$$\Rightarrow \frac{C_s Q^2}{2} = \frac{M^2}{2} (C_h + C_s) + DC_o$$

$$\Rightarrow Q^2 = M^2 \left(\frac{C_h + C_s}{C_s} \right) + \frac{2DC_o}{C_s} \quad \dots(2)$$

and

$$\frac{d}{dM}(TC) = \frac{MC_h}{Q} - \frac{(Q - M)}{Q} C_s = 0$$

$$\Rightarrow \frac{M}{Q} (C_h + C_s) - C_s = 0$$

$$\Rightarrow M = \left(\frac{C_s}{C_h + C_s} \right) Q \quad \dots(3)$$

Substituting the value of M in equation (2), we get

$$Q^2 = \left(\frac{C_s}{C_h + C_s} \right)^2 \times \left(\frac{C_h + C_s}{C_s} \right) Q^2 + \frac{2DC_o}{C_s}$$

$$Q^2 \left[1 - \frac{C_s}{C_h + C_s} \right] = \frac{2DC_o}{C_s}$$

$$\Rightarrow Q^2 \left[1 - \frac{C_s}{C_h + C_s} \right] = \frac{2DC_o}{C_s}$$

$$\Rightarrow Q^2 \left(\frac{C_h}{C_h + C_s} \right) = \frac{2DC_o}{C_s}$$

$$\Rightarrow Q^2 = \frac{2DC_o}{C_h} \left(\frac{C_s + C_h}{C_s} \right)$$

Thus, the optimum value of Q is given by

$$Q^* = \sqrt{\frac{2DC_o}{C_h} \left(\frac{C_s + C_h}{C_s} \right)}$$

The optimum value of M is given by

$$M^* = \left(\frac{C_s}{C_h + C_s} \right) Q^* = \sqrt{\frac{2DC_o}{C_h} \left(\frac{C_s}{C_h + C_s} \right)}$$

Therefore, the minimum inventory cost

$$TC^* = \sqrt{2DC_o C_h \left(\frac{C_s}{C_h + C_s} \right)}$$

$$\text{Total Cycle time} = t^* = \frac{Q^*}{D} = \sqrt{\frac{2C_o}{DC_h} \left(\frac{C_s + C_h}{C_s} \right)}$$

Let us explain this model with the help of an example.

Example 2: A contractor has to supply engines to a truck manufacturer at the rate of 20 per day. The penalty in the contract is `100 per engine per day for missing the scheduled delivery date. The cost of holding an engine in stock for one month is `150. His production process is such that each month (30 days) he starts producing a batch of engines through the agencies and all are available for supply after the end of the month. Determine the maximum inventory level at the beginning of each month.

Solution: Here we have

$$C_h = \frac{150}{30} \text{ per engine per day} = `5 \text{ per engine per day};$$

$$C_s = `100 \text{ per day per engine}; D = 20 \text{ engines per day and } t^* = 30 \text{ days.}$$

The maximum inventory level M^* at the beginning of each month will be

$$M^* = \frac{C_s}{C_h + C_s} Q^* = \frac{C_s}{C_h + C_s} (D \times t^*)$$

$$= \frac{100}{5 + 100} \times 20 \times 30 = 571 \text{ engines}$$

You may now like to solve the following problem to assess your understanding.

- E3)** A manufacturer has to supply his customer with 24000 units of his product every year. The demand is fixed and known. Since the unit is used by the customer in an assembly operation and the customer has no storage space for units, if the manufacturer fails to supply the required units, the shortage cost is `2 per unit per month. The inventory cost is `1 per unit per month and the set-up cost per run is `3500. Determine
- the optimal run size (Q);
 - the optimal level of inventory (M) at the beginning of any period;
 - the optimal scheduling period; and
 - the minimum inventory cost.

8.6 EOQ MODEL WITH UNIFORM REPLENISHMENT

For this model, it is assumed that the production run may take significant time to complete, that is, the replenishment is not instantaneous. The stock is added at a constant rate, i.e., the replenishment is uniform. Other assumptions are the same as the ones made for the EOQ model with uniform demand as explained in Sec. 8.3. The order size here is taken as the production size.

The graph representing this situation is shown in Fig. 8.4

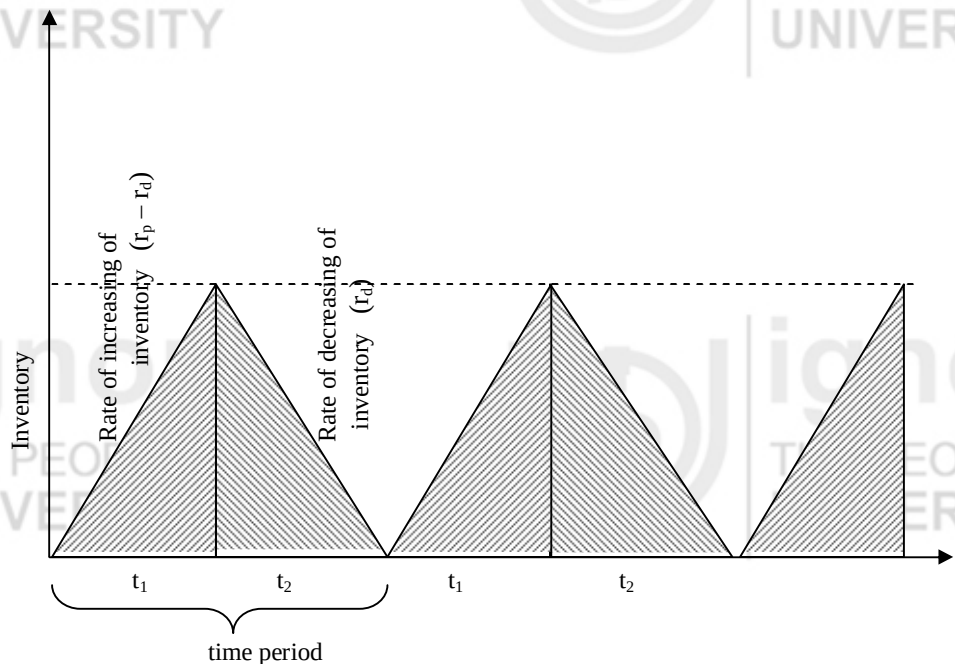


Fig. 8.4

Let r_d be the demand rate in units per time period, r_p , the replenishment rate per unit of time, t_1 , the time of production run or time of replenishment, and t_2 , the time required for the inventory to be exhausted. Then $t_1 + t_2 = t$ (say) is the time for each order cycle.

Let Q be the number of units produced per order cycle. Then

$$t_1 = \frac{Q}{r_p} \quad (\because \text{Time for producing 1 unit} = 1/\text{Rate of replenishment} = 1/r_p)$$

Inventory is building up at the rate of $r_p - r_d$.

$$\therefore \text{Maximum inventory level} = (r_p - r_d) t_1$$

$$\text{Average inventory} = \frac{(r_p - r_d) t_1}{2} = \frac{(r_p - r_d)}{2} \frac{Q}{r_p} = \frac{Q}{2} \left(1 - \frac{r_d}{r_p} \right)$$

$$\text{Now, ordering (or set-up) cost} = \frac{D}{C} C_o \text{ and carrying cost} = \frac{Q}{2} \left(1 - \frac{r_d}{r_p} \right) C_h$$

$\therefore$ the total annual inventory cost is given by

$$TC = \frac{D}{C} C_o + \frac{Q}{2} \left(1 - \frac{r_d}{r_p} \right) C_h$$

$$\text{The cost will be minimum if } \frac{D}{C} C_o = \frac{Q}{2} \left(1 - \frac{r_d}{r_p} \right) C_h$$

$$\Rightarrow Q^2 = \frac{2DC_o}{C_h} \left(\frac{r_p}{r_p - r_d} \right) \Rightarrow Q^* = \sqrt{\frac{2DC_o}{C_h} \left(\frac{r_p}{r_p - r_d} \right)}$$

Thus, the characteristics of the model are as follows:

1. Optimum number of production runs per year:

$$N^* = \frac{D}{Q^*} = \sqrt{\frac{DC_h (r_p - r_d)}{2C_o r_p}}$$

2. Length of each lot size production run:

$$t_1 = \frac{Q^*}{r_p} = \sqrt{\frac{2DC_o}{C_h r_p (r_p - r_d)}}$$

3. The total minimum inventory cost:

$$TC^* = \frac{D}{Q^*} C_o + \frac{Q^*}{2} \left(1 - \frac{r_d}{r_p} \right) C_h = \sqrt{2DC_o C_h \left(1 - \frac{r_d}{r_p} \right)}$$

We take up an example to apply this model.

Example 3: A tyre producer makes 1600 tyres per day and sells them at approximately half that rate. Accounting figures show that the production

set-up cost is ₹1000 and carrying cost per unit is ₹5. If the annual demand is 160000 tyres, what is the optimal lot size and how many production runs should be scheduled per year?

Solution: We are given that the annual demand (D) = 160000 tyres, C_h = ₹5, C_o = ₹1000, Production rate (r_p) = 1600 tyres per day and Demand rate (r_d) = 800 tyres per day.

$$\therefore \text{The optimal lot size } Q^* = \sqrt{\frac{2DC_o}{C_h} \times \frac{r_p}{r_p - r_d}}$$

$$Q^* = \sqrt{\frac{2 \times 160000 \times 1000}{5} \times \frac{1600}{1600 - 800}} = 11314 \text{ tyres}$$

$$\text{Optimal production runs per year } N^* = \frac{D}{Q^*} = \frac{160000}{11314} = 14 \text{ runs/year approx.}$$

You may now like to solve the following problems to assess your understanding.

- E4)** A product is manufactured at the rate of 50 items per day. The demand occurs at the rate of 30 items per day. It is given that the set-up cost per order is ₹1000 and holding cost per unit time is ₹0.05. Find the economic lot size and the associated total cost per cycle assuming that no shortage is allowed.
- E5)** A company uses 100000 units of a particular item per year. Each item costs ₹2. The production engineering department estimates the holding cost as 12.5% of the value of the inventory per day. The replenishment rate is uniform at 500 units per day. Assuming 250 working days (for replenishment purpose), calculate the
- optimal set-up quantity;
 - total cost on the basis of optimal policy; and
 - optimal number of set-ups.

8.7 EOQ MODEL WITH PRICE (OR QUANTITY) DISCOUNTS

If the items are purchased in bulk, some discount in price is usually offered by the supplier. The discount is called **all units discount** if it is applicable for all the units purchased. It is said to be **incremental discount** if discounts are offered only for the items which are in excess of the specified amount. In the incremental discount, the prices offered in different slabs are applicable in finding the total cost while in 'all units discount', only one price at any one slab is applicable for finding the total cost. Buying in large quantities may result in the following advantages:

Less unit price, less ordering costs, less transportation cost, fewer stock-outs and hence less chances of loss of goodwill, mass display by retailers and preferential treatment by the sellers.

At the same time, large quantity buying may involve the following disadvantages:

Higher carrying costs, older stocks, lower stock turnover, more capital required, blocked money, less flexibility, heavier deterioration and depreciation.

If the items are purchased with quantity discounts, the price C may vary according to the following scheme:

$$\begin{aligned}
 C &= p_0 \text{ if purchased quantity } Q = Q_0 < b_1 \\
 &= p_1 \text{ if purchased quantity } b_1 \leq Q_1 < b_2 \\
 &\dots\dots\dots \\
 &\dots\dots\dots \\
 &\dots\dots\dots \\
 &= p_n \text{ if purchased quantity } b_n < Q_n
 \end{aligned}$$

where p_{j-1} is greater than p_j , for $j = 1, 2, \dots, n$. Here p_j is the price per unit for the j^{th} lot size. If the shortages are not allowed and supply is instantaneous, the total cost per year is given by the set of relations:

$$TC(Q_j) = Dp_j + \frac{C_0 D}{Q_j} + \frac{1}{2} C_h Q_j, \text{ where } b_j \leq Q_j < b_{j+1}$$

and $C_h = ip_j$, where i is the percentage change for $j = 0, 1, 2, \dots, n$.

We follow the procedure given below to find the overall optimum lot size:

Step 1: We find EOQ for the lowest price, i.e., we calculate

$$Q_n^* = \sqrt{\frac{2C_0 D}{ip_n}}$$

If $Q_n^* \geq b_n$, the optimum order quantity is Q_n^* . If $Q_n^* < b_n$, we go to Step 2.

Step 2: We compute $Q_{n-1}^* = \sqrt{\frac{2C_0 D}{ip_{n-1}}}$ for the next lowest price.

If $Q_{n-1}^* \geq b_{n-1}$, then we compare the total cost TC_{n-1} for purchasing Q_{n-1}^* with the total cost TC_n for purchasing quantity b_n and select the one that gives the least cost.

If $Q_{n-1}^* < b_{n-1}$, we go to Step 3.

Step 3: We compute $Q_{n-2}^* = \sqrt{\frac{2C_0 D}{ip_{n-2}}}$. If $Q_{n-2}^* \geq b_{n-2}$, we compare the total cost TC_{n-2} , TC_{n-1} and TC_n for purchase of quantities Q_{n-2}^* , b_{n-1} and b_n , respectively, and select the optimum purchase quantity.

If $Q_{n-2}^* < b_{n-2}$, then we go to Step 4.

Step 4: We continue in this fashion until $Q_{n-j}^* \geq b_{n-j}$. Then we compare total costs TC_{n-j} with $TC_{n-j+1}, \dots, TC_{n-1}, TC_n$ for purchase quantities Q_{n-j}^* , $b_{n-j+1}, \dots, b_{n-1}, b_n$, respectively, and select the optimum purchase quantity.

We now apply this method to an example.

Example 5: The annual demand for an item is 2400 units. The inventory carrying charge is 24% of the purchase price per year. Purchase prices are:

$$p_1 = ₹ 10 \quad \text{for purchasing } Q_1 < 500$$

$$p_2 = ₹ 9.25 \quad \text{for purchasing } 500 \leq Q_2 < 750$$

$$p_3 = ₹ 8.75 \quad \text{for purchasing } 750 \leq Q_3$$

Determine the optimum purchase quantity taking the ordering cost as

$$\text{i) } ₹ 437.5$$

$$\text{ii) } ₹ 87.5$$

Solution: To determine the optimum purchase quantity, we follow the procedure given below:

$$\text{i) As per Step 1, } Q_3^* = \sqrt{\frac{2 \times ₹ 437.5 \times 2400}{(0.24)(₹ 8.75)}} = 1000 \text{ units}$$

Since 1000 is greater than 750, the optimum purchase quantity is 1000 units.

$$\text{ii) Here } Q_3^* = \sqrt{\frac{2 \times ₹ 87.5 \times 2400}{(0.24)(₹ 8.75)}} = 447 \text{ units}$$

Since $447 < 750 = b_3$, we next compute

$$Q_2^* = \sqrt{\frac{2 \times ₹ 87.5 \times 2400}{(0.24)(₹ 9.25)}} = 435 \text{ units}$$

Since $435 < 500 = b_2$, we next compute

$$Q_1^* = \sqrt{\frac{2 \times ₹ 87.5 \times 2400}{(0.24)(₹ 10)}} = 418 \text{ units} > 0 \text{ and } < 500$$

We now compare the total cost for purchasing $Q_1^* = 418$, $b_2 = 500$ and $b_3 = 750$ units, respectively.

$$\begin{aligned} TC_1(\text{for purchasing } 418) &= ₹ 10 \times 2400 + \frac{₹ 87.5 \times 2400}{418} + \frac{1}{2}(0.24) \times (₹ 10) \times 418 \\ &= ₹ 24504 \end{aligned}$$

$$\begin{aligned} TC_2(\text{for } Q_2 = 500) &= ₹ 9.25 \times 2400 + \frac{₹ 87.5 \times 2400}{500} + \frac{1}{2}(0.24) \times (₹ 9.25) \times (500) \\ &= ₹ 23175 \end{aligned}$$

$$\begin{aligned} TC_3(\text{for } Q_3 = 750) &= ₹ 8.75 \times 2400 + \frac{₹ 87.5 \times 2400}{750} + \frac{1}{2}(0.24) \times (₹ 8.75) \times (750) \\ &= ₹ 22067 \end{aligned}$$

As the total inventory cost is minimum for $Q_3 = 750$, therefore, the optimum purchase quantity is 750.

You may now like to solve the following problems to assess your understanding.

E6) Consider Example 5 and determine the optimum purchase quantity with the ordering cost as ₹ 175.

8.8 SUMMARY

1. An inventory is a physical stock of idle resources of any kind having some economic value kept for the purpose of meeting future demand. It indicates the raw material required before production, the finished goods after production ready for delivery to consumers, human resources, financial resources, etc. stocked in order to meet an expected demand in the future.
2. Inventories are essential for almost all businesses for efficient and smooth operations. Hence, inventories need to be controlled in the most profitable way.
3. The quantity that is kept in stock in order to balance the costs of holding too much stock vis-à-vis the costs of ordering in small quantities is called the economic order quantity.
4. Five models have been discussed for obtaining the economic order quantity for the following situations:
 - i) when demand is uniform,
 - ii) when rates of demand are different in different cycles,
 - iii) when shortages are allowed,
 - iv) when replenishment is uniform, and
 - v) when price discounts are given.

8.9 SOLUTIONS/ANSWERS

E1) We have $D = \frac{12000}{1.25} = \frac{12000}{125} \times 100 = 9600$; $C = 1.25$ per unit;

$$C_o = 15 \text{ and } C_h = 15\% \text{ of } 1.25 \Rightarrow \frac{15}{100} \times \frac{125}{100}$$

$$Q^* = \sqrt{\frac{2DC_o}{C_h}} = \sqrt{\frac{2 \times 9600 \times 15 \times 100}{15}} = \sqrt{1920000} = 1385 \text{ units}$$

E2) Here $C_o = 36$, $C = 2$, $D = 10000$, and $C_h = 18\% \text{ of } 2 = \frac{18 \times 2}{100} = \frac{9}{25}$

$$\text{Therefore, a) } Q^* = \sqrt{\frac{2 \times 10000 \times 36 \times 25}{9}} = \sqrt{2000000} = 1414 \text{ units}$$

$$\text{and b) } \frac{D}{Q^*} = \frac{10000}{1414} = 7.072$$

$$\text{Thus, the optimum no. of days} = \frac{365}{7} = 52 \text{ days}$$

E3) We have $D = \text{number of units supplied per year} = 24000$

Ordering cost per unit = ₹ 3500

Shortage cost = ₹ 2 per unit per month

= ₹ 24 per unit per year

Carrying cost = ₹ 1 per unit per month

= ₹ 12 per unit per year

i) The optimal value of Q is

$$\begin{aligned} Q^* &= \sqrt{\frac{2DC_o}{C_h} \left(\frac{C_s + C_h}{C_s} \right)} \\ &= \sqrt{\frac{2 \times 24000 \times 3500}{12} \left(\frac{24 + 12}{24} \right)} \\ &= \sqrt{2 \times 1000 \times 3500 \times 3} = \sqrt{210 \times 100000} \\ &= \sqrt{21} \times 1000 = 4.583 \times 1000 = 4583 \text{ units} \end{aligned}$$

$$\begin{aligned} \text{ii) } M^* &= \left(\frac{C_s}{C_h + C_s} \right) Q^* = \left(\frac{24}{12 + 24} \right) \times 4583 \\ &= \frac{2}{3} \times 4583 = 3055 \text{ units} \end{aligned}$$

$$\text{iii) } t^* = \frac{Q^*}{D} = \frac{4583}{24000} = \frac{4.583}{2} = 2.29 \text{ months}$$

iv) The minimum inventory cost = TC*

$$\begin{aligned} &= \sqrt{2DC_o C_h \left(\frac{C_s}{C_h + C_s} \right)} \\ &= \sqrt{2 \times 24000 \times 12 \times \left(\frac{24}{12 + 24} \right) \times 3500} \\ &= 576000 \times \frac{24}{36} = 37200 \end{aligned}$$

E4) We have D = 30 items per day, $r_p = 50$ items per day,

$C_o = 1000$, $C_h = 0.05$, $r_d = 30$ items per day

$$\begin{aligned} \text{Therefore, } Q^* &= \sqrt{\frac{2DC_o}{C_h} \left(\frac{r_p}{r_p - r_d} \right)} = \sqrt{\frac{2 \times 30 \times 1000}{0.05} \left(\frac{50}{20} \right)} \\ &= \sqrt{3000000} = 1732.05081 = 1732 \text{ units} \end{aligned}$$

Total minimum inventory cost

$$\begin{aligned} TC^* &= \sqrt{2DC_o C_h \left(1 - \frac{r_d}{r_p} \right)} \\ &= \sqrt{2 \times 30 \times 1000 \times 0.05 \left(1 - \frac{30}{50} \right)} \\ &= \sqrt{2 \times 30 \times 1000 \times \frac{5}{100} \times \frac{20}{50}} = \sqrt{1200} = 34.64 \end{aligned}$$

E5) We are given that

$r_p = 500 \times 250$ units per year, $r_d = 100000$ units per year, $C_o = 25$

$$C_h = 12.5\% \text{ of } 2 = \frac{12.5 \times 2}{100} = \frac{1}{4} = 0.25$$

$$\begin{aligned} \text{a) } Q^* &= \sqrt{\frac{2DC_o}{C_h} \left(\frac{r_p}{r_p - r_d} \right)} \\ &= \sqrt{\frac{2 \times 100000 \times 25}{0.25} \left(\frac{125000}{25000} \right)} \\ &= \sqrt{4000000 \times 25} = 10000 \end{aligned}$$

$$\begin{aligned} \text{b) } TC^* &= \sqrt{2DC_o C_h \left(1 - \frac{r_d}{r_p} \right)} \\ &= \sqrt{2 \times 100000 \times 0.25 \times 25 \left(1 - \frac{100000}{125000} \right)} \\ &= \sqrt{250000} = 500 \end{aligned}$$

$$\begin{aligned} \text{Total cost on the basis of optimal policy} &= TC^* + 2 \times 100000 \\ &= 200500 \end{aligned}$$

$$\text{c) Optimum number of set ups} = \frac{D}{Q^*} = \frac{100000}{10000} = 10 \text{ set-ups}$$

$$\text{E6) Here } Q_3^* = \sqrt{\frac{2 \times 175 \times 2400}{(.24)(8.75)}} = 632 \text{ units}$$

Since $632 < 750 = b_3$, we next compute

$$Q_2^* = \left(\frac{2 \times 175 \times 2400}{(.24)(9.25)} \right)^{1/2} = 615 \text{ units}$$

Since $615 > 500 = b_2$ we compare the total costs for purchasing $Q_2^* = 615$ and $b_3 = 750$ units respectively.

$$\begin{aligned} TC_2(\text{for } 615 \text{ units}) &= 9.25 \times 2400 + \frac{100 \times 2400}{615} + \frac{1}{2} \times (0.24) \times (9.25) \times (615) \\ &= ₹ 23,273 \end{aligned}$$

$$\begin{aligned} TC_3(\text{for } Q_3 = 750) &= 8.75 \times 2400 + \frac{100 \times 2400}{750} + \frac{1}{2} \times (0.24) \times (8.75) \times (750) \\ &= ₹ 22,119.50 \end{aligned}$$

Therefore, the economic purchase quantity for this problem is

$$Q_3^* = 750 \text{ units}$$

Role of inventory in services

1. Ensuring Service Quality and Customer Satisfaction:

- **Availability of Materials and Supplies:**

Inventory ensures that service providers have the necessary materials, tools, and equipment readily available to complete jobs and meet customer needs, says Elmasys.

- **Reducing Delays and Improving Efficiency:**

Well-managed inventory minimizes delays caused by stockouts or shortages, allowing for faster service delivery and increased efficiency.

- **Building Trust and Loyalty:**

Consistent, reliable service, supported by efficient inventory management, builds trust and loyalty among customers.

2. Enhancing Operational Efficiency and Cost Savings:

- **Streamlined Processes:**

Inventory management systems can automate tasks like tracking inventory levels, placing orders, and managing product movement, streamlining operations.

- **Cost Optimization:**

By reducing waste, minimizing storage costs, and optimizing purchasing decisions, inventory management can help service businesses save money.

- **Improved Decision-Making:**

Real-time inventory data provides insights into demand patterns, sales trends, and stock levels, enabling more informed decision-making.

3. Supporting Forecasting and Planning:

- **Demand Forecasting:**

Accurate inventory data helps service businesses forecast future demand for materials and supplies, allowing for better planning and resource allocation.

- **Budgeting and Financial Planning:**

Inventory management plays a crucial role in budgeting and financial planning, providing insights into costs and potential savings.

4. Examples of Inventory in Service Businesses:

- **Field Service Businesses:**

Inventory management is critical for field service companies to ensure technicians have the necessary parts and supplies to complete repairs and installations.

- **Restaurants and Food Service:**

Inventory management ensures restaurants have enough ingredients, supplies, and equipment to meet customer demand and maintain quality standards.

- **Healthcare:**

Hospitals and clinics rely on efficient inventory management to track pharmaceuticals, medical supplies, and equipment.

- **Professional Services:**

Even businesses like consulting firms and accounting practices may need to inventory supplies like software licenses, templates, and legal documents, says Gross Mendelsohn.

Supply Chain

A supply chain is a network of entities, resources, and activities involved in bringing a product or service from raw materials to the end-user. It includes everything from sourcing materials, manufacturing, and transporting products to retail and customer fulfillment. Effective supply chain management aims to improve efficiency, reduce costs, and enhance customer satisfaction.

Key Components of a Supply Chain

- **Sourcing/Procurement:** Acquiring raw materials and components from suppliers.
- **Manufacturing/Production:** Transforming raw materials into finished products.
- **Distribution:** Moving finished goods from production to customers, often involving warehouses, transportation, and retail channels.
- **Reverse Logistics:** Handling returns, repairs, and recycling of products.

Entities Involved in a Supply Chain:

- **Suppliers:** Provide raw materials or components.
- **Manufacturers:** Produce goods.
- **Transporters/Logistics Providers:** Move goods between different points in the chain.
- **Warehouses/Distribution Centers:** Store and manage inventory.
- **Retailers:** Sell products to consumers.
- **Customers:** The end users of products or services.

Supply Chain Management (SCM)

SCM is the process of planning, organizing, directing, and controlling the flow of goods and services within the supply chain. It aims to optimize efficiency, reduce costs, and improve customer satisfaction.

Managing suppliers of service

Managing suppliers of service notes effectively involves a strategic approach encompassing clear communication, performance tracking, and building strong relationships. This includes defining clear expectations, regularly monitoring performance, and proactively managing risks associated with service notes.

Here's a breakdown of key strategies:

1. Establish Clear Expectations and Communication:

- **Define roles and responsibilities:**

Ensure both parties understand their obligations and responsibilities regarding service note content, delivery, and quality.

- **Maintain open communication channels:**

Foster regular communication to facilitate prompt resolution of issues and ensure alignment on expectations.

- **Utilize a formal contract:**

Formalize agreements with suppliers, including service levels, key performance indicators (KPIs), and dispute resolution mechanisms.

2. Track Performance and Monitor Service Notes:

- **Implement KPIs:**

Develop and track KPIs to measure supplier performance, such as service note accuracy, delivery time, and response time.

- **Regularly review performance:**

Conduct regular reviews of service note quality and supplier performance against established KPIs.

- **Identify and address issues proactively:**

Use data and feedback to identify potential issues and implement corrective actions promptly.

3. Manage Risk and Build Strong Relationships:

- **Assess and mitigate risks:**

Identify potential risks associated with service note content, delivery, and data security.

- **Diversify supplier base:**

Consider diversifying suppliers to reduce dependence on a single source and minimize risk.

- **Foster collaboration and trust:**

Build strong relationships with suppliers through open communication, collaboration, and a mutual commitment to achieving shared goals.

4. Leverage Technology and Tools:

- **Utilize supplier management software:**

Implement software solutions to streamline supplier management processes, track performance, and facilitate communication.

- **Embrace digital tools:**

Utilize digital platforms to improve communication, collaboration, and information sharing with suppliers.

5. Continuously Improve and Adapt:

- **Seek feedback and iterate:**

Regularly solicit feedback from suppliers and internal stakeholders to identify areas for improvement and refine service note processes.

- **Adapt to changing needs:**

Stay informed about evolving industry trends and adapt service note processes to meet changing business needs.

By implementing these strategies, organizations can effectively manage their service note suppliers, build strong relationships, and optimize their operations.

The Vehicle Routing Problem (VRP) involves finding the most efficient routes for a fleet of vehicles to visit a set of customers, aiming to minimize costs, travel time, or other objectives while adhering to constraints like capacity or time windows. It's a generalization of the Traveling Salesperson Problem (TSP) where a single vehicle visits all locations. VRPs are widely used in logistics, transportation, and delivery services.

Key Concepts:

- **Depot:** The starting and ending point for all vehicles.
- **Customers:** Locations that need to be visited by vehicles.
- **Vehicles:** The means of transportation used to deliver goods or services.
- **Routes:** The specific paths that vehicles take to visit customers.
- **Constraints:** Restrictions on the routes, such as capacity limits, time windows, or other operational requirements.
- **Objective:** The goal of the optimization, such as minimizing total distance, travel time, or number of vehicles used.

Types of VRPs:

- **Basic VRP:**

Focuses on minimizing total travel distance without time windows or capacity constraints.

- **Capacitated VRP (CVRP):**

Each vehicle has a limited capacity, and the objective is to minimize total distance while adhering to capacity constraints.

- **Vehicle Routing Problem with Time Windows (VRPTW):**

Each customer has a specific time window within which they must be visited.

- **Other variants:**

VRPs can be further customized to include other constraints such as pick-up and delivery, split deliveries, or multiple depots.

Why is VRP important?

- **Efficiency:**

VRP solutions help optimize fleet utilization and reduce costs.

- **Delivery Time:**

Optimized routes can lead to faster deliveries and better customer satisfaction.

- **Sustainability:**

By reducing mileage and the number of vehicles, VRPs can contribute to lower carbon emissions and environmental impact.

- **Logistics Optimization:**

VRPs are crucial for managing complex logistics operations, such as delivery and distribution networks.

Solving VRPs:

- **Heuristic algorithms:**

Methods like simulated annealing, genetic algorithms, and tabu search are used to find near-optimal solutions for large and complex VRPs.

- **Exact algorithms:**

Methods like branch-and-cut can find optimal solutions for smaller instances of VRPs.

- **Software and tools:**

Various software and online platforms offer VRP solving capabilities, often using advanced algorithms and heuristics.